Interoperability Maturity Lab (IML)

Founding Manuscript

A Scientific Framework for Trusted Health Information Ecosystems

Status: Open for Scientific Review

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Master Index of the Manuscript

This single numbered index replaces previous mixed index/register structures. It is designed as a navigation table and a thematic compatibility check for the current manuscript.

The index covers all major themes developed in the text: health before technology, trustworthy information, Health Information Ecosystems, interoperability layers, the IML assessment framework, methodology, AMR/BMR as demonstrator, transition from assessment to action, Open Clinical Workspace, responsible AI, roadmap and annexes.

No.	Manuscript section	Themes and content checked against the text
06	**Part IV - The IML Framework**	Translates the concepts into six assessment domains: Governance and Standards, Technical Interoperability, Identity/Consent/Trust, Adoption and Use, Security and Resilience, Feedback/Correction/Learning, with Institutional Engagement, Payer Interoperability, AI Readiness, professional role alignment and institutional responsiveness as cross-cutting concerns.
07	**Part V - Methodology**	Describes indicators, evidence grading, scoring philosophy, country profiles rather than rankings, complementarity with the Global Digital Health Monitor (GDHM) and existing digital health maturity dashboards, weighting, data sources, expert review, validation, versioning, reproducibility and methodological limitations.
08	**Part VI - Demonstrator: Antimicrobial Resistance and Multidrug-Resistant Bacteria**	Uses AMR/BMR as the first operational thread of IML. It focuses on resistant isolate versus clinically meaningful infection, UTI and MDR E. coli, minimum dataset, microbiology, clinical interpretation, treatment, outcomes, payers, AI and public health learning.
09	**Part VII - From Assessment to Action**	Shows how assessment becomes practical improvement through action logic, maturity profiles, support for low-resource ecosystems, high-resource fragmentation, the French paradox, the Danish reference without idealisation, institutional inertia, institutional responsiveness, community building, software-quality benchmarking and pilot-to-service pathways.
10	**Part VIII - Open Clinical Workspace**	Proposes a vendor-neutral implementation bridge that interacts with existing systems through import on demand, open standards, modular architecture, a clinical context layer, identity and consent, audit, analytics, database quality, operating-system quality, professional role support, digital health software quality benchmarking and AI assistance under human responsibility.
11	**Part IX - Responsible Artificial Intelligence and Scientific Integrity**	Defines AI as an ecosystem actor under human responsibility, with transparency of AI-assisted writing, ethical inspiration from the OpenAI Charter, privacy, proportionality, bias, equity, reproducibility, AI readiness and the boundaries of automation.
12	**Part X - Roadmap and Future Work**	Sets out the pathway from manuscript to working programme: versioning, external scientific review, methodology handbook, technical handbook, case studies, implementation pathway, open-source reference implementation, publication/DOI/open documentation, community building and future research agenda.

No.	Manuscript section	Themes and content checked against the text
13	**Annexes**	Provides operational working instruments: glossary, acronyms, indicator catalogue, evidence grading, AMR/BMR minimum dataset, payer interoperability indicators, institutional engagement/readiness indicators, thematic coherence review, change log, concept-to-literature mapping and suggested citation/review template.

How to Read this Manuscript

Established evidence

Statements that should eventually be supported by peer-reviewed literature, international recommendations or official documentation.

IML proposal

Concepts or methodological choices proposed by IML for discussion and refinement.

Research hypothesis

Ideas that appear plausible and useful but require further validation.

Future work

Sections, indicators or case studies that remain to be developed.

Opening Statement

Health is humanity's common objective.

"Health is our purpose. Trustworthy information is our foundation. Interoperability is our path."

Progress in health information should not be measured by the sophistication of technology alone, but by its capacity to help people make better decisions, reduce suffering and improve health together.

Every day, billions of clinical observations, laboratory analyses, medical images, prescriptions, epidemiological reports and scientific discoveries contribute to our understanding of health.

Yet information alone does not protect health.

Information protects health only when it becomes trusted knowledge, available to the right person, at the right time, under the right conditions.

This transformation cannot be achieved by technology alone; it requires scientific knowledge, institutions, governance, trust and, above all, collaboration.

The Interoperability Maturity Lab (IML) was created to support this ambition.

This manuscript proposes a scientific framework for understanding, assessing and progressively improving Health Information Ecosystems.

It is not intended as a definitive answer, but as an invitation to build, together, a future in which trustworthy information contributes more effectively to protecting health.

Preface

Why this manuscript?

Healthcare has entered an unprecedented digital era. Across the world, hospitals, primary care practices, laboratories, pharmacies, insurers, public health authorities and research institutions generate and exchange vast quantities of health information every day. Yet despite remarkable technological progress, healthcare professionals, patients and decision-makers continue to face a familiar challenge: information often remains fragmented when it is most needed.

The consequences are tangible. Clinical decisions may be made with incomplete information. Diagnostic procedures may be unnecessarily repeated. Prevention opportunities may be missed. Public health surveillance may identify resistant microorganisms without fully understanding the clinical situations in which they occur. Even in highly digitalised health systems, the continuity of information frequently remains more difficult to achieve than the continuity of care itself.

This observation is not a criticism of existing initiatives. On the contrary, the remarkable achievements of the World Health Organization, HL7 International, SNOMED International, openEHR, LOINC, IHE, national health authorities and many other organisations have transformed healthcare over recent decades. Their work constitutes the essential foundation upon which future progress must continue to build.

Complementarity with the Global Digital Health Monitor and digital health maturity initiatives

IML is also complementary to the Global Digital Health Monitor (GDHM) and to other digital health maturity initiatives. GDHM provides a valuable country-level view of digital health capacity and progress. IML does not seek to duplicate this work. Its contribution is narrower and more clinically oriented: to examine whether digital health capacity preserves meaning, context, trust and usefulness across Health Information Ecosystems.

A second distinction is important for scientific interpretation. The WHO-hosted GDHM beta platform indicates that it will replace the existing platform once the data on the site have been reviewed and validated by participating Member States. This means that GDHM country information should be read with attention to its validation status. During preparation of this version, publicly visible government-approved country-level validation appeared limited, with Austria identified as the only clearly government-validated country entry in the materials reviewed. This observation should be treated as time-sensitive and rechecked in later versions.

This limitation reinforces the complementarity of IML. GDHM provides a valuable high-level monitoring structure, but IML can go further where validated information is scarce, incomplete or insufficiently clinical: it can document the evidence level, identify uncertainty, invite external review and test interoperability through concrete use cases such as AMR/BMR rather than relying only on national self-description.

However, an important question remains: how can health information better protect health? This manuscript was written to explore that question.

The Interoperability Maturity Lab (IML) does not propose another technical standard, another electronic health record or another digital platform. Instead, it proposes a scientific and practical framework for understanding, assessing and progressively improving Health Information Ecosystems.

The central hypothesis of this work is simple.

Health is the objective. Information is the foundation. Interoperability is the means.

Digital technologies have no intrinsic value unless they contribute to improving health. Likewise, interoperability should never be regarded as an objective in itself. Its purpose is to enable trustworthy information to circulate safely, meaningfully and responsibly wherever it is required to support prevention, diagnosis, treatment, rehabilitation, research and public health.

Throughout this manuscript, health is understood according to the holistic definition adopted by the World Health Organization: a state of physical, mental and social well-being rather than merely the absence of disease. Such a definition naturally extends beyond healthcare institutions and invites us to consider health information as part of a much broader ecosystem involving professionals, patients, laboratories, public authorities, researchers, public and private payers, industry and society as a whole.

For this reason, the concept of a Health Information Ecosystem occupies a central place in this work. The objective is no longer simply to evaluate individual information systems, but to understand how an entire ecosystem contributes to protecting health through trustworthy information.

This manuscript also recognises that many challenges cannot be solved by technology alone. Governance, institutional engagement, professional collaboration, public trust, legal frameworks and education all influence the maturity of health information ecosystems. Technical interoperability, while indispensable, represents only one component of a much broader transformation.

IML therefore adopts an open, collaborative and evidence-informed approach. It seeks neither to replace existing initiatives nor to rank countries. Instead, it proposes a practical roadmap that can support continuous improvement while respecting the diversity of health systems throughout the world.

The first practical application proposed in this manuscript focuses on antimicrobial resistance and multidrug-resistant bacteria. This choice reflects the global importance of antimicrobial resistance as a public health priority and illustrates how meaningful clinical decisions depend upon the integration of microbiological, clinical and epidemiological information rather than isolated datasets.

Finally, this manuscript recognises the emergence of artificial intelligence as a powerful tool capable of assisting health professionals, researchers and policymakers. The preparation of this work benefited from the editorial and analytical support of OpenAI's ChatGPT. In accordance with the aspiration expressed in the OpenAI Charter that advanced artificial intelligence should "benefit all of humanity," the use of AI within IML is understood as an instrument supporting scientific reflection rather than replacing human responsibility. All scientific interpretations, methodological choices and conclusions presented in this manuscript remain the responsibility of the author.

This is not the description of a completed system or the presentation of a finished methodology; it is an invitation.

An invitation to researchers, clinicians, engineers, policymakers, patients and institutions to contribute to the progressive development of trustworthy, interoperable and human-centred Health Information Ecosystems.

The future of healthcare will not be determined solely by the technologies we build.

It will be determined by our collective ability to transform health information into knowledge that genuinely protects health.

The IML Declaration

A Declaration for Health Information Ecosystems

Health is one of humanity's greatest shared responsibilities. Every healthcare professional, patient, researcher, institution and policymaker contributes, directly or indirectly, to its protection.

Over the past decades, remarkable progress has been achieved through advances in medicine, public health, digital technologies and biomedical research. Yet despite these achievements, health information remains fragmented across organisations, professions and institutional boundaries. Too often, the information required to support prevention, diagnosis, treatment, research or public health is incomplete, delayed or inaccessible when it is most needed.

The Interoperability Maturity Lab (IML) was created in response to this challenge.

IML is an independent scientific initiative dedicated to understanding, assessing and progressively improving Health Information Ecosystems. It does not promote a particular software platform, technical standard, commercial product or national model. Instead, it seeks to provide an open scientific framework capable of supporting continuous improvement through evidence, collaboration and practical implementation.

IML considers interoperability not as an objective in itself but as one of the essential conditions required to protect health. Digital technologies acquire value only when they contribute to better prevention, safer clinical decisions, improved continuity of care, stronger public health and more equitable access to trustworthy information.

The framework recognises that technical interoperability alone is insufficient. Sustainable progress also depends upon governance, institutional engagement, semantic consistency, legal certainty, professional collaboration, public trust and education. Health Information Ecosystems therefore extend beyond software and infrastructures to include the human, organisational and societal dimensions of healthcare.

IML further recognises that health systems differ considerably across countries. Economic resources, institutional structures, legal frameworks and digital maturity vary widely. For this reason, the purpose of IML is not to rank countries or identify a single ideal model. Its objective is to support each health system in progressing from its current situation towards greater interoperability, trust and effectiveness.

Particular attention is given to the continuity of health information. Continuity of care cannot be fully achieved without continuity of trustworthy information across healthcare providers, laboratories, public health authorities, public and private payers, research organisations and patients themselves.

The framework also recognises that health financing organisations are integral components of Health Information Ecosystems. Public health insurance systems, national health services and private insurers all contribute information that may improve prevention, care coordination, chronic disease management and health system performance when appropriate governance and interoperability are achieved.

Artificial intelligence represents another emerging opportunity. IML recognises AI as a powerful instrument capable of assisting professionals, researchers and policymakers while maintaining that scientific judgement, ethical responsibility and clinical decision-making remain fundamentally human responsibilities. The preparation of this manuscript benefited from the editorial and analytical assistance of OpenAI's ChatGPT. In accordance with the aspiration expressed in the OpenAI Charter that advanced artificial intelligence should "benefit all of humanity," AI is understood within IML as an instrument serving science and society rather than replacing human responsibility.

Finally, IML recognises that no institution, country or organisation can achieve interoperability alone. Progress depends upon openness, cooperation and continuous learning. Every ecosystem can improve. Every country can contribute. Every experience represents an opportunity to strengthen collective knowledge.

This manuscript therefore constitutes neither a finished methodology nor a definitive solution.

It is an invitation.

An invitation to researchers, clinicians, engineers, health authorities, public and private payers, software developers, patients and policymakers to participate in the progressive construction of trustworthy Health Information Ecosystems that better protect health.

The IML Commitment

The Interoperability Maturity Lab is committed to:

It is committed to putting health before technology, promoting trustworthy health information, supporting prevention through better information continuity, encouraging open standards and scientific collaboration, respecting the diversity of national health systems, evaluating ecosystems rather than isolated technologies, promoting responsible and human-centred artificial intelligence, supporting continuous improvement rather than competition between countries, recognising public and private payers as integral actors of Health Information Ecosystems, and contributing to better health for present and future generations through trustworthy, interoperable and ethically governed health information.

PART 0

Why This White Paper Exists

"The purpose of this manuscript is not to introduce another digital health initiative. It is to understand how health information can better protect health."

0.1 Why another white paper?

Healthcare is experiencing one of the most profound transformations in its history. Digital technologies have fundamentally changed the production, storage and exchange of health information. Electronic health records, laboratory information systems, medical imaging, genomics, artificial intelligence and national digital health infrastructures have become integral components of modern healthcare.

These developments have been accompanied by extraordinary international efforts. Organizations such as the World Health Organization (WHO), HL7 International, SNOMED International, openEHR, IHE, LOINC, the OECD, the European Commission and many national agencies have established standards, governance models and implementation frameworks that continue to shape digital health worldwide.

This manuscript does not question the importance of these achievements; on the contrary, it is built upon them.

However, despite this remarkable progress, a fundamental observation remains.

Health information often remains fragmented when it is most needed: clinical decisions may still rely on incomplete information, patients may repeat investigations that already exist elsewhere, healthcare professionals may spend valuable time searching for information instead of using it, researchers may struggle to access structured clinical data, and public health authorities may receive microbiological findings without sufficient clinical context.

The problem is therefore no longer simply digitalisation, but transformation: how can information become trusted knowledge that genuinely contributes to protecting health? This question constitutes the starting point of the Interoperability Maturity Lab.

0.2 A changing paradigm

For many years, digital health focused primarily on computerising healthcare; later, the emphasis shifted towards interoperability, and today a new transition is emerging, from merely exchanging information to enabling information to improve decisions.

This represents a profound paradigm shift.

Information acquires value not because it exists, but because it contributes to better prevention, safer diagnosis, more appropriate treatment and stronger public health.

Technology therefore becomes an instrument rather than an objective, while health remains the objective.

0.3 Why Health Information Ecosystems?

Healthcare is frequently described through individual information systems, including hospital information systems, electronic medical records, laboratory information systems, radiology information systems and pharmacy systems, each of which fulfils an important function.

However, health rarely follows organisational boundaries: patients move, professionals collaborate, samples circulate and knowledge evolves.

Consequently, healthcare should no longer be analysed as a collection of isolated information systems.

It should be understood as a Health Information Ecosystem.

Within such ecosystems, patients, professionals, laboratories, public health authorities, public and private payers, researchers, industry and policymakers all contribute to producing, interpreting and using health information.

This concept forms the conceptual foundation of IML.

0.4 Beyond technical interoperability

Interoperability is frequently described as a technical capability. Technical, semantic and organisational interoperability are all essential, yet practical experience demonstrates that these dimensions alone do not guarantee meaningful collaboration.

Successful interoperability also depends upon institutional commitment, governance, legislation, education, professional trust and sustainable cooperation.

IML therefore proposes extending the discussion beyond technology.

The maturity of a Health Information Ecosystem depends upon both technical and institutional readiness.

0.5 A practical scientific framework

IML does not seek to replace existing standards.

It seeks to organise them within a coherent framework oriented towards health outcomes.

Its ambition is practical.

Every maturity assessment should answer three practical questions: where are we today, what should improve next, and how will progress be measured?

Assessment therefore becomes an instrument for continuous improvement rather than comparison.

0.6 The role of public and private payers

One dimension remains surprisingly underrepresented in most digital health maturity frameworks.

Health financing organisations, public health insurance systems, national health services, private insurers and mutual organisations possess essential information concerning healthcare utilisation, continuity of care, long-term follow-up and prevention.

Yet information exchange between public and private payers often remains limited or fragmented.

In many countries, public reimbursement systems and complementary private insurance organisations operate independently despite serving the same patients.

IML considers payer interoperability an essential component of Health Information Ecosystems.

Its evaluation should therefore become part of maturity assessment.

0.7 Why antimicrobial resistance?

Every scientific framework requires practical validation.

IML proposes antimicrobial resistance, and particularly multidrug-resistant bacteria, as its first demonstrator.

AMR illustrates the limitations of fragmented information particularly well.

Microbiological data alone cannot distinguish colonisation from infection.

Antibiograms alone cannot determine clinical relevance.

Meaningful surveillance requires integration of laboratory results, symptoms, diagnoses, treatments, outcomes and epidemiological context.

AMR therefore represents an ideal use case for demonstrating how interoperability contributes directly to better health.

0.8 A framework for every country

Countries differ considerably: some possess highly developed digital infrastructures, others continue to face major resource constraints, some rely predominantly on public financing and others combine public and private systems. IML does not assume that one model fits all.

Instead, it proposes a progressive methodology capable of supporting continuous improvement according to national priorities, available resources and institutional contexts.

The objective is not perfection, but progress.

0.9 Complementarity with the Global Digital Health Monitor and digital health maturity initiatives

The Global Digital Health Monitor (GDHM) and other digital health maturity tools help describe national digital health capacity, including leadership, governance, strategy, investment, legislation, services, infrastructure, standards, interoperability and workforce. They provide useful high-level views for planning, comparison and policy dialogue.

The validation status of GDHM data should also be acknowledged. The WHO beta version states that its data are to be reviewed and validated by participating Member States before replacing the existing platform. In the public materials reviewed for this manuscript version, government-approved country-level validation appeared limited, with Austria identified as the only clearly government-validated country entry. This should not be read as a weakness of GDHM, but as a reminder that national maturity monitoring depends on official validation cycles and may therefore remain incomplete or delayed.

IML can add value precisely in this interval. It can provide an evidence-graded complementary layer that records uncertainty, distinguishes documented evidence from hypothesis, and evaluates whether digital health capacity is clinically useful in concrete pathways. In this sense, IML does not copy GDHM; it extends the question from national digital health maturity to the practical usefulness of interoperability for care, public health, research and learning.

IML addresses a different question. It does not primarily ask whether digital health capacity exists. It asks whether this capacity allows trustworthy health information to remain clinically meaningful, contextualised, auditable and actionable across the ecosystem.

This distinction is important. A country may have a national strategy, digital services, infrastructure and standards, while still struggling to preserve clinical context, link laboratory information to diagnosis and treatment, distinguish resistant isolates from clinically meaningful infections, support correction and feedback, or transform information into shared knowledge.

IML therefore proposes a complementary layer of analysis. Existing maturity tools describe digital health capacity. IML examines interoperability usefulness: whether information can support better care, prevention, public health, research, payer coordination and responsible artificial intelligence.

0.10 Towards a shared future

Health is increasingly global.

Pandemics, antimicrobial resistance, ageing populations, chronic diseases and artificial intelligence all demonstrate that health challenges rarely respect institutional or national boundaries.

The future therefore depends not only upon better technologies but also upon stronger cooperation.

IML is proposed as a contribution to this collective effort.

It does not claim to possess all the answers.

It proposes a framework within which better questions may be asked, better evidence generated and better decisions supported.

Closing Statement

The future of healthcare will not be determined by the quantity of information we produce, but by our collective ability to transform trustworthy information into knowledge that protects health.

PART I

Health Before Technology

"Technology changes rapidly. Health remains humanity's enduring objective."

1.1 Beginning with Health

Most publications on digital health begin by describing technologies, software platforms, interoperability standards or digital infrastructures. This manuscript deliberately adopts a different perspective.

The purpose of healthcare is not to build digital systems. Its purpose is to protect and improve health.

Digital technologies therefore represent instruments rather than objectives.

Electronic health records, laboratory information systems, artificial intelligence and interoperability standards acquire value only when they contribute to better prevention, safer diagnosis, more effective treatment, stronger public health and improved health outcomes.

For this reason, the Interoperability Maturity Lab begins not with technology, but with health.

This choice is more than editorial.

It defines the entire conceptual framework presented throughout this manuscript.

1.2 The WHO Definition of Health

The Constitution of the World Health Organization defines health as:

"Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity."

Although formulated in 1946, this definition remains remarkably relevant.

It reminds us that health extends beyond biological disease.

Health includes psychological well-being, social participation, living conditions, environmental influences, education, economic stability and access to healthcare.

Consequently, no single healthcare institution can completely represent an individual's health.

Likewise, no isolated information system can adequately describe it.

Health is inherently multidimensional.

Its information must therefore also be multidimensional.

1.3 Health as a Dynamic Continuum

Health is often perceived as a binary distinction between healthy and ill, but reality is considerably more complex.

Individuals continuously move along a dynamic continuum influenced by ageing, behaviour, genetics, environmental exposure, healthcare interventions and social determinants.

This continuum begins long before disease appears, continues throughout life, and makes the objective of healthcare not merely the treatment of disease but the prevention of unnecessary deterioration.

This observation fundamentally changes the role of health information: it should not merely document disease, but help prevent disease.

1.4 Prevention and Early Detection

One of the greatest opportunities offered by modern health information ecosystems lies in prevention.

Most serious diseases develop progressively, and before diagnosis weak signals may already exist, including laboratory abnormalities, repeated consultations, medication patterns, hospital admissions, microbiological findings and lifestyle indicators; individually these observations may appear insignificant, but collectively they may represent valuable clinical knowledge.

Unfortunately, they often remain distributed across multiple organisations.

Consequently, prevention frequently fails not because information does not exist, but because information cannot be integrated meaningfully.

1.5 Information as a Determinant of Health

Traditional determinants of health include:

This manuscript proposes that trustworthy health information deserves recognition as another enabling determinant.

Information itself does not improve health, but trustworthy information enables better decisions, better decisions improve prevention, and better prevention improves health.

The quality of health information therefore indirectly influences health outcomes.

1.6 Continuity of Care and Continuity of Information

Continuity of care has long been recognised as a fundamental principle of high-quality healthcare.

Patients benefit when professionals possess sufficient knowledge of previous diagnoses, treatments, preferences and outcomes.

However, continuity of care depends upon something even more fundamental.

Continuity of trustworthy information.

A patient's journey frequently involves multiple healthcare organisations, including primary care, hospitals, laboratories, pharmacies, home care, public health authorities and public or private payers; each interaction generates valuable information.

If this information cannot accompany the patient appropriately, continuity of care becomes fragmented.

For this reason, IML introduces the concept of Continuity of Health Information.

1.7 From Data to Knowledge

Modern healthcare produces extraordinary quantities of data, yet data alone rarely improve health. Data become useful only after interpretation: data become information, information becomes knowledge, knowledge supports clinical decision-making, and clinical decisions influence health outcomes.

The objective of interoperability is therefore not merely exchanging data, but enabling trustworthy knowledge.

1.8 Health Information Ecosystems

Healthcare information is generated by numerous participants, including patients, healthcare professionals, hospitals, primary care, laboratories, pharmacies, public health authorities, universities, research organisations, public and private insurers, industry and artificial intelligence systems.

Together, these actors constitute a Health Information Ecosystem whose performance cannot be understood by evaluating isolated software applications, because its maturity depends upon relationships.

Those relationships involve governance, trust, standards, institutional collaboration, semantic consistency, clinical relevance and human expertise.

1.9 Why Technology Alone Is Not Enough

Technological progress remains essential.

However, experience demonstrates that technology alone rarely transforms healthcare.

Successful digital transformation also requires:

Interoperability therefore represents a socio-technical challenge rather than a purely technical one.

1.10 Towards Health Information Ecosystems

This chapter has deliberately begun with health rather than technology.

Health remains the objective, information provides the foundation, and interoperability constitutes one of the practical conditions required to transform information into trustworthy knowledge capable of supporting better prevention, safer clinical decisions and stronger public health.

The following chapters therefore move beyond individual information systems towards the broader concept of Health Information Ecosystems, which forms the conceptual core of the Interoperability Maturity Lab.

Key Messages

At the conclusion of this first part, several principles emerge: health precedes technology, prevention requires trustworthy information, continuity of care depends upon continuity of health information, Health Information Ecosystems constitute the appropriate unit of analysis, and interoperability is a means of protecting health rather than an objective in itself.

1.11 Health Information and Health Equity

Health information is not distributed equally.

Some individuals benefit from highly integrated healthcare systems in which clinical histories, laboratory results, imaging studies and prescriptions are readily available to healthcare professionals.

Others receive care across fragmented organisations where information remains incomplete, delayed or inaccessible.

This disparity contributes to inequalities in healthcare itself.

Health equity therefore depends not only on access to healthcare services but also on access to trustworthy health information.

Individuals living with chronic diseases, multiple comorbidities, disabilities or complex social situations frequently interact with numerous healthcare providers throughout their lives.

The more fragmented these interactions become, the greater the risk of incomplete clinical understanding.

Consequently, information fragmentation may amplify existing social and health inequalities.

Improving Health Information Ecosystems therefore represents not only a technological objective but also an opportunity to promote greater equity in healthcare.

This perspective aligns closely with the World Health Organization's commitment to universal health coverage and equitable access to quality healthcare.

IML consequently considers equity as an implicit outcome of mature Health Information Ecosystems.

The objective is not simply to connect systems.

It is to reduce inequalities created by fragmented information.

1.12 Information Across the Entire Health Journey

Health information does not begin when disease appears and does not end when treatment finishes. It accompanies every stage of life, including health promotion, prevention, screening, primary care, hospital care, rehabilitation, long-term follow-up, palliative care, public health, research and health financing; each stage generates information, contributes knowledge and influences future decisions.

The challenge therefore extends beyond continuity of care. It concerns the continuity of health information throughout an individual's entire health journey. For this reason, IML introduces the concept of the Health Information Lifecycle.

The lifecycle begins with the generation of information. It continues through validation, interpretation, sharing, reuse, preservation and responsible governance.

Artificial intelligence increasingly participates in several of these stages, but AI does not replace professional judgement. Its contribution depends entirely upon the quality, completeness and trustworthiness of the underlying information.

Consequently, trustworthy Health Information Ecosystems become essential not only for healthcare professionals but also for the safe and responsible use of artificial intelligence.

1.13 Health Information and Public Trust

No Health Information Ecosystem can function without trust: patients must trust that their information will be handled responsibly, healthcare professionals must trust the information they receive, institutions must trust each other's governance mechanisms, researchers must trust data quality and public authorities must trust surveillance systems.

Trust therefore becomes an invisible infrastructure supporting interoperability.Technical excellence alone cannot compensate for insufficient trust. Conversely, trust without reliable information cannot support evidence-based healthcare.

IML therefore considers trust both as a prerequisite and as an outcome of mature Health Information Ecosystems.

1.14 Towards a New Vision of Interoperability

The traditional question has often been whether systems can exchange data. IML proposes a different question: can this ecosystem transform trustworthy information into knowledge that better protects health?

This change of perspective has profound consequences.

The unit of analysis becomes the ecosystem rather than the application, the objective becomes health rather than technology, and the measure of success becomes improved decision-making rather than the quantity of exchanged data.

Interoperability therefore evolves from a technical capability into a strategic component of healthcare quality, prevention and societal resilience.

Closing Reflection

Throughout history, medicine has advanced through scientific discovery, professional collaboration and public trust.

Digital technologies now provide unprecedented opportunities to strengthen these foundations.

Their true value, however, will never be measured by the number of connected systems, the volume of exchanged data or the sophistication of software platforms.

Their value will be measured by a simpler question: do they help people remain healthier? This question guides every chapter that follows.

PART II

Health Information Ecosystems

2.1 From Information Systems to Health Information Ecosystems

Healthcare has undergone one of the most significant technological transformations in modern history. Over recent decades, hospitals, primary care practices, laboratories, pharmacies, imaging centres and public health agencies have progressively adopted sophisticated information systems capable of capturing, storing and processing unprecedented quantities of digital information. These developments have transformed clinical practice, facilitated biomedical research and strengthened public health surveillance.

Despite these achievements, an important limitation has progressively become apparent. Healthcare is still frequently organised around institutions, whereas health itself follows people.

Patients move continuously between healthcare providers. A patient may receive vaccinations in primary care, emergency treatment in a hospital, laboratory investigations in an independent laboratory, rehabilitation in another institution and long-term follow-up through community services. Older adults living with chronic diseases often receive care simultaneously from general practitioners, specialists, hospitals, home-care teams, pharmacies, social services and multiple financing organisations.

Each interaction generates valuable health information. Each organisation develops legitimate responsibilities regarding the information it produces.

However, no organisation possesses the complete picture.

The consequence is a paradox. Healthcare has become increasingly digital while clinical knowledge frequently remains fragmented. Information exists, but it often remains distributed across institutional boundaries that reflect administrative structures rather than the patient's actual journey.

This observation requires a profound conceptual change.

Rather than considering healthcare as a collection of independent information systems, IML proposes understanding it as a Health Information Ecosystem.

The ecosystem becomes the fundamental unit of analysis.

2.2 Defining the Health Information Ecosystem

The concept of ecosystem has been widely adopted in environmental sciences, economics and innovation studies to describe dynamic systems composed of multiple interacting actors whose collective behaviour cannot be understood by analysing each component independently.

Healthcare presents similar characteristics.

Clinical decisions emerge from interactions between patients, professionals, laboratories, public health authorities, financing organisations, researchers, regulators and increasingly artificial intelligence systems.

For this reason, IML proposes the following definition.

A Health Information Ecosystem is the dynamic network of individuals, organisations, technologies, governance mechanisms and information flows that collectively generate, validate, exchange, interpret and use trustworthy health information in order to protect and improve health.

Several elements distinguish this definition.

First, the ecosystem is dynamic rather than static. Scientific knowledge evolves continuously. Clinical practice changes. Regulations develop. New technologies emerge. The ecosystem therefore represents an adaptive system rather than a fixed architecture.

Second, information alone is insufficient.

Information acquires value only when its quality, provenance, integrity, semantic consistency and clinical relevance permit trustworthy interpretation.

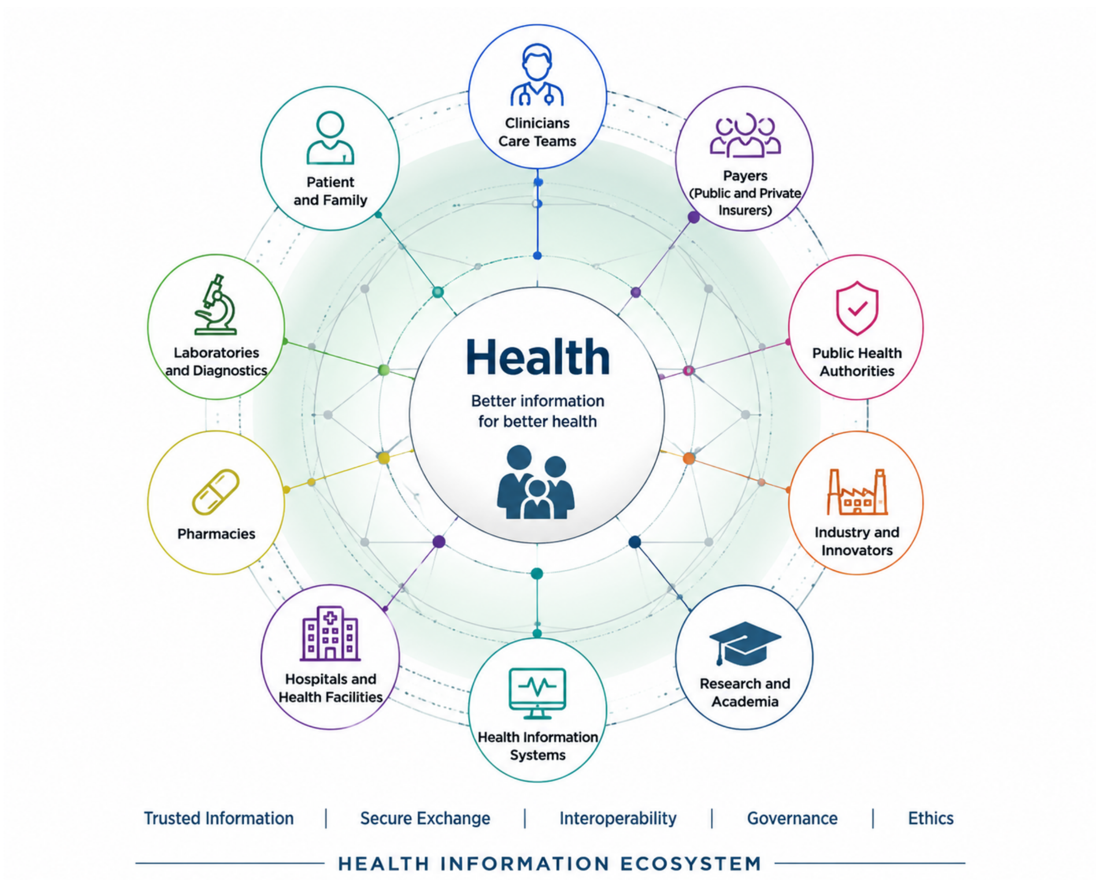
Third, the ecosystem exists for a single purpose.

Its objective is neither technological performance nor institutional optimisation.

Its objective is health.

This distinction fundamentally differentiates IML from technology-centred approaches to interoperability.

Figure 2.1 - Health Information Ecosystem (inserted).



2.3 The Actors of the Ecosystem

Every Health Information Ecosystem is composed of participants whose responsibilities differ but whose objectives converge.

Patients are not passive recipients of healthcare. They produce information through symptoms, experiences, behaviours, preferences and outcomes. Increasingly, they also generate information through connected devices, patient-reported outcome measures and personal health applications.

Healthcare professionals transform observations into clinical reasoning. General practitioners, specialists, nurses, pharmacists and allied health professionals continuously enrich the ecosystem through diagnosis, treatment decisions and longitudinal follow-up.

Laboratories contribute biological evidence that frequently determines diagnosis, therapeutic choices and epidemiological surveillance. Their information acquires full meaning only when interpreted within the patient's clinical context.

Hospitals integrate multidisciplinary expertise, coordinate complex interventions and generate large volumes of structured and unstructured information.

Public health authorities observe populations rather than individuals. Their role consists of identifying trends, emerging risks and opportunities for prevention while respecting ethical and legal frameworks governing personal information.

Universities and research organisations transform routine clinical information into scientific knowledge capable of improving future healthcare.

Public insurers, national health services and private payers occupy a unique position within the ecosystem. Their information frequently extends over many years and multiple providers, offering longitudinal perspectives that are rarely available within individual healthcare institutions. For this reason, IML recognises payer interoperability as an essential dimension of ecosystem maturity rather than a secondary administrative function.

Industry contributes medicines, diagnostic technologies, medical devices and digital innovations. Within mature ecosystems, industrial innovation should strengthen rather than fragment information continuity.

Artificial intelligence represents the newest participant in the ecosystem. AI neither replaces clinicians nor constitutes an autonomous decision-maker. Instead, it functions as a knowledge amplifier whose usefulness depends entirely upon the quality and trustworthiness of the information available.

No participant alone possesses complete knowledge.

Only the ecosystem can progressively approach a comprehensive understanding of health.

2.3.1 Professional roles, qualification and digital transformation

Health Information Ecosystems do not depend only on software, standards or institutional governance. They also depend on the way health professions are trained, authorised, recognised and coordinated.

This dimension deserves explicit attention. In many health systems, the formal structure of health professional education has changed more slowly than the technological, organisational and informational environment in which care is now delivered. Medical education remains long, demanding and socially valued, but much of this training was designed for a model of practice in which clinical authority, diagnosis and decision-making were concentrated around the physician. Contemporary healthcare increasingly requires additional competencies: information stewardship, teamwork, data interpretation, digital literacy, quality improvement, public health reasoning, patient communication, shared decision-making, cybersecurity awareness and responsible use of artificial intelligence.

At the same time, nurses, pharmacists, laboratory professionals, allied health professionals, care coordinators, data managers, public health workers and other health professions have progressively taken a larger role in real healthcare activity. They contribute to prevention, chronic disease follow-up, medication safety, diagnostic pathways, patient education, surveillance, quality improvement and digital workflows. However, in many ecosystems, professional authority, remuneration structures, symbolic hierarchy and legal responsibility continue to reflect older models in which medical dominance remains largely intact.

This mismatch creates a governance problem for interoperability. If the real work of care is distributed across many professions, but decision rights, access rights, data-entry responsibilities, correction mechanisms and accountability remain organised around a narrow professional hierarchy, health information may become distorted. Some actors who produce essential information may lack recognition or digital access. Others may carry formal responsibility without receiving the training, tools or institutional support required for modern information-intensive care.

IML therefore proposes that maturity assessment should include professional role alignment. The question is not whether one profession should replace another. The question is whether each profession’s training, legal scope, digital access, responsibility, remuneration and contribution to information quality are aligned with the real work required to protect health.

This issue is particularly relevant in digital health. A health system may invest heavily in electronic records, AI tools, dashboards and interoperability standards while leaving professional education and role distribution largely unchanged. Such a system risks creating advanced digital infrastructure around outdated professional boundaries. Digital transformation then becomes an additional burden rather than a genuine improvement.

For IML, the quality of a Health Information Ecosystem depends on whether professionals are prepared and authorised to contribute to trustworthy information. This includes the ability to enter meaningful data, interpret information from other sources, correct errors, participate in shared governance, use digital tools safely, understand the limits of AI and explain information use to patients.

Professional hierarchy should therefore be assessed not as a political accusation, but as an ecosystem maturity question. Mature systems should be able to ask whether medical, nursing, pharmacy, laboratory, public health, administrative and technical roles are appropriately distributed, whether training reflects current practice, whether remuneration and recognition support useful work, and whether all relevant actors can contribute to knowledge continuity.

In this sense, workforce transformation is part of interoperability. Information cannot become trustworthy shared knowledge if the people who generate, interpret, correct and act upon it are trained for one system, governed by another, and working inside a third.

2.4 Relationships Rather than Organisations

Traditional health information strategies frequently evaluate organisations independently: hospitals evaluate hospitals, laboratories evaluate laboratories, and primary care evaluates primary care.

Yet patients experience healthcare differently, through transitions between hospital and home, specialist and general practitioner, diagnosis and treatment, acute care and prevention.

Every transition represents an information relationship.

The maturity of a Health Information Ecosystem therefore depends less upon the quality of its individual components than upon the quality of the relationships connecting them.

Relationships become measurable.

They may be evaluated through continuity, trust, semantic consistency, governance, institutional commitment and clinical usefulness.

This perspective represents one of the central conceptual contributions of IML.

2.5 Information Flows: From Data Production to Shared Knowledge

Every Health Information Ecosystem is fundamentally sustained by information flows.

Information does not remain static. It is continuously generated, transformed, interpreted, enriched and reused throughout the health journey.

Traditional information systems frequently focus on data storage.

Health Information Ecosystems focus on information movement.

This distinction is fundamental.

A laboratory information system may successfully produce a microbiological result.

However, its contribution to health depends entirely upon the subsequent journey of that information.

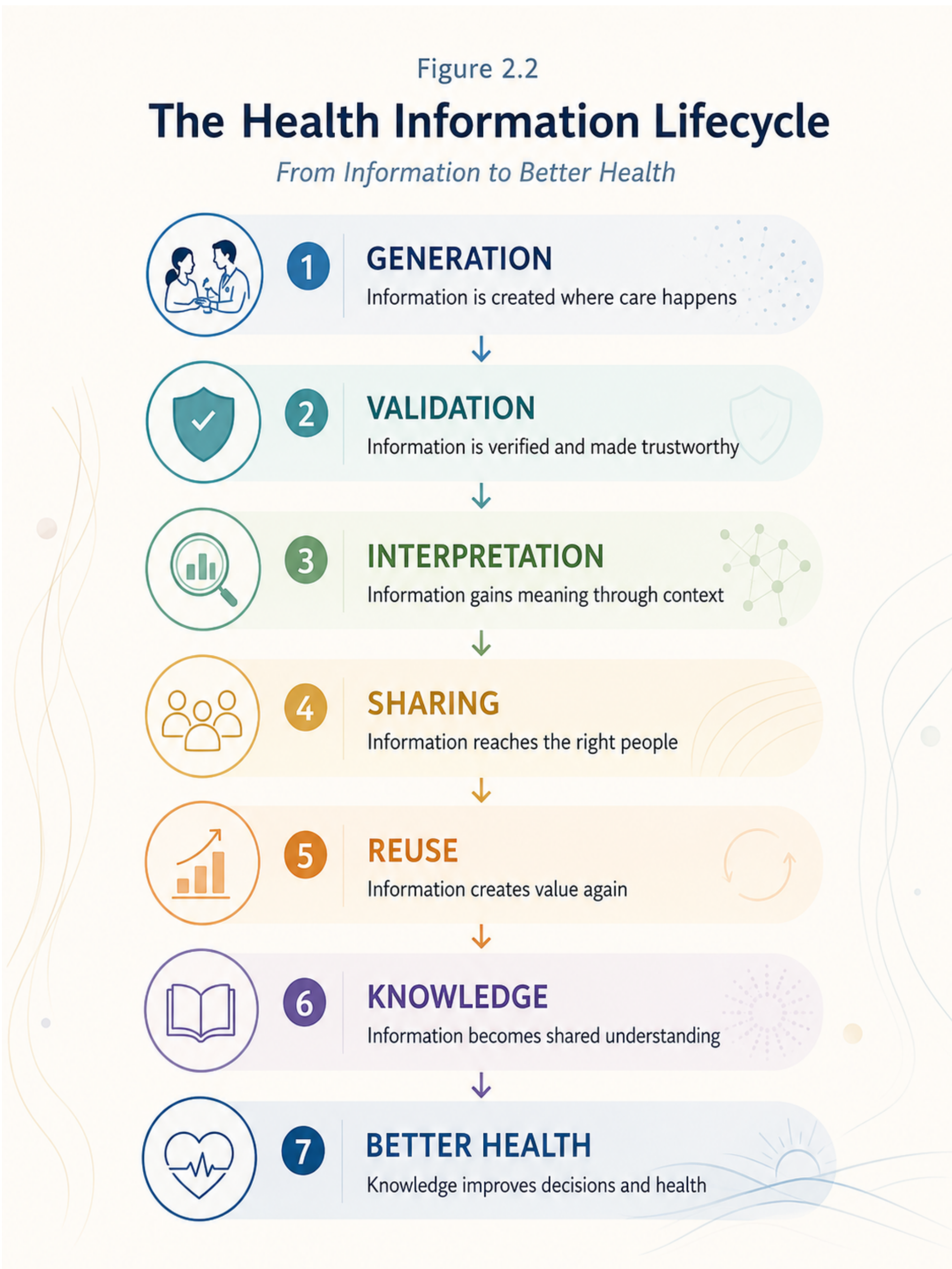
Its value depends on whether the clinician receives it promptly, whether it can be interpreted within the patient's clinical context, whether infection control teams can use it, whether surveillance systems can identify emerging outbreaks, and whether researchers can reuse anonymised information to improve future practice.

Information therefore acquires value through circulation rather than accumulation.

2.5.1 The Health Information Lifecycle

IML proposes describing information as a continuous lifecycle rather than a static resource.

The lifecycle includes generation, validation, interpretation, sharing, reuse and preservation, each of which contributes to the transformation of isolated data into usable knowledge.



Generation refers to information originating from clinical encounters, laboratory analyses, medical imaging, prescriptions, administrative procedures, patient-reported outcomes, public health surveillance and increasingly connected devices.

Validation makes information trustworthy through clinical validation, laboratory quality assurance, professional accountability and semantic consistency.

Interpretation gives data meaning; a laboratory result, for example, acquires significance only when integrated with symptoms, history, examination findings and epidemiological context.

Sharing means that trustworthy information becomes available to authorised participants through clearly defined governance mechanisms; it does not imply unrestricted access, but appropriate access.

Reuse allows health information to acquire value beyond its original purpose, serving clinical care, public health, research, quality improvement, education, health system planning and responsible artificial intelligence.

Preservation recognises that some information retains value for decades, particularly longitudinal clinical histories, vaccination records, occupational exposures, rare diseases and family history; future generations of professionals may benefit from decisions made today.

2.5.2 Information Has Context

One of the greatest limitations of fragmented digital health systems lies in the loss of context.

A laboratory result without symptoms, a diagnosis without microbiology, a prescription without indication or a hospital discharge without follow-up may lose meaning when separated from its clinical environment.

For this reason, IML introduces the concept of Clinical Context Preservation.

Interoperability should preserve not only information but also the context that gives information its meaning.

2.5.3 Information Quality

Health Information Ecosystems depend upon trustworthy information.

Trustworthiness results from multiple dimensions.

Trustworthiness depends on accuracy, completeness, timeliness, consistency, traceability, clinical relevance, governance, security and semantic interoperability.

None of these dimensions alone guarantees quality.

Together, however, they determine whether information deserves professional trust.

2.5.4 From Information to Shared Knowledge

Healthcare has traditionally measured information exchange, but IML proposes measuring something more demanding: the creation of shared knowledge.

Knowledge emerges when multiple actors contribute complementary perspectives.

A microbiologist contributes laboratory evidence, a general practitioner contributes longitudinal history, a specialist contributes diagnostic expertise, a pharmacist contributes medication safety, a public insurer contributes continuity of care, a researcher contributes scientific interpretation and artificial intelligence contributes pattern recognition.

Together, these perspectives generate knowledge that none of the participants could produce independently.

This collaborative production of knowledge represents one of the defining characteristics of mature Health Information Ecosystems.

2.5.5 Knowledge Continuity

Perhaps the most important consequence of interoperability is not data continuity, or even information continuity alone, but Knowledge Continuity.

Knowledge should accompany both patients and healthcare systems throughout time.

Every clinical encounter should enrich future decision-making rather than beginning from zero.

Every generation of professionals should inherit trustworthy knowledge from previous generations.

Every research project should contribute to improving future healthcare.

Knowledge continuity therefore represents one of the highest objectives of mature Health Information Ecosystems.

2.6 Governance of Health Information Ecosystems

A Health Information Ecosystem cannot be reduced to a technical architecture. It is also a governed space in which rights, responsibilities, priorities and limits must be defined. Governance determines who may create information, who may validate it, who may access it, who may correct it and under which conditions it may be reused.

In fragmented systems, governance is often distributed across multiple institutions. A hospital governs its own records. A laboratory governs its own results. A payer governs reimbursement information. A public health authority governs surveillance data. Each governance perimeter may be legitimate, yet the patient journey crosses all of them.

The central governance challenge is therefore not the existence of local responsibility. Local responsibility is necessary. The challenge is the absence of shared mechanisms that allow information to move across boundaries without losing integrity, meaning, accountability or trust.

IML proposes that ecosystem governance should be assessed through several dimensions: clarity of responsibilities, transparency of access rules, mechanisms for correction, patient rights, professional accountability, data provenance, ethical oversight, cybersecurity, auditability and public legitimacy.

A mature ecosystem does not maximise information circulation. It governs it. The objective is not unrestricted sharing, but appropriate access to trustworthy information for legitimate health purposes. This distinction is essential. Interoperability without governance may threaten trust. Governance without interoperability may preserve control while preventing useful care, research and public health action.

The role of governance is therefore to organise responsible circulation. It must protect privacy while enabling care. It must prevent misuse while supporting prevention. It must preserve institutional accountability while allowing collaboration. This balance is one of the defining challenges of modern health information ecosystems.

2.7 Trust as Infrastructure

Trust is often treated as an ethical value. In Health Information Ecosystems, it also functions as infrastructure. Without trust, information does not circulate, professionals hesitate to rely on external data, patients resist sharing, institutions protect their silos and public health systems lose legitimacy.

Trust has several layers. Patients must trust that their information will be used responsibly. Clinicians must trust that information received from another system is accurate, complete and clinically relevant. Laboratories must trust that their results will not be detached from necessary interpretive context. Public health authorities must trust data quality. Researchers must trust provenance and reproducibility. Institutions must trust each other's governance mechanisms.

Technical security contributes to trust, but it does not exhaust it. Encryption, authentication and access control are indispensable, yet trust also depends upon transparency, accountability, professional culture, legal clarity and the possibility of correction. A technically secure system may still be distrusted if its governance is opaque or its purposes are poorly understood.

IML therefore treats trust as both a prerequisite and an outcome of maturity. It is a prerequisite because interoperability requires willingness to rely on information produced elsewhere. It is an outcome because well-governed, transparent and clinically useful information exchange can progressively strengthen confidence among participants.

This approach leads to a practical question for every maturity assessment: not only can information be exchanged, but can it be trusted by those who must use it?

2.8 Institutional Engagement

Technical standards are necessary, but they do not implement themselves. Health information ecosystems depend upon institutions that are willing and able to participate in sustained collaboration. This is the domain of Institutional Engagement.

Institutional Engagement refers to the observable capacity of organisations to enter dialogue, share expertise, participate in common projects, respond to external initiatives, contribute to governance and support practical implementation. It does not imply agreement with every proposal. Rather, it reflects a willingness to engage with the collective problem of information continuity.

The development of IML suggests that this dimension deserves explicit attention. Many digital health initiatives possess technical logic, scientific relevance or public health value, yet progress slowly because the institutional conditions for collaboration are weak. Contact points may be unclear. Responsibilities may be dispersed. Incentives may be misaligned. Participation may depend on individual goodwill rather than formal pathways.

This observation should not be interpreted as criticism of specific organisations or countries. It is a research hypothesis: interoperability maturity may depend not only on standards, infrastructure and legal frameworks, but also on the practical responsiveness and collaborative behaviour of institutions.

For this reason, IML proposes that Institutional Engagement be considered a cross-cutting component of ecosystem maturity. It may be assessed through the existence of identified contact points, transparent collaboration procedures, participation in national or international networks, openness to external review, publication of methodology, involvement of clinicians and researchers, and the ability to support pilot projects.

Institutional Engagement should also include responsiveness to external scientific and operational proposals. In digital health, the absence of a reply, contact pathway or review mechanism is not a neutral administrative detail. It may indicate that the ecosystem lacks the institutional interface required to transform useful ideas into evaluated action.

This is especially important when health information challenges evolve faster than traditional administrative cycles. An ecosystem may have strategies, standards and digital programmes while still lacking the practical capacity to receive, triage, evaluate or redirect external proposals. IML therefore treats institutional responsiveness as a maturity capability: the ability to answer, examine, connect or openly decline proposals through transparent procedures.

2.9 Institutional Readiness

Institutional Engagement describes willingness to participate. Institutional Readiness describes the capacity to do so effectively.

An institution may support interoperability in principle while lacking the practical conditions required for implementation. It may lack technical staff, legal procedures, data governance processes, budget, leadership support or operational time. In such situations, collaboration remains aspirational even when the institutional intention is genuine.

Institutional Readiness therefore concerns preparedness. It includes organisational structures, decision-making pathways, data protection expertise, interoperability skills, project management capacity, funding mechanisms and the ability to evaluate outcomes.

IML distinguishes readiness from maturity. A highly mature system may still be unready for a specific collaboration if governance is unclear. A less mature system may be highly ready to progress if leadership, incentives and operational capacity are aligned.

This distinction is important for countries with different resources. Low-resource environments may display strong readiness but limited infrastructure. High-resource environments may possess advanced infrastructure but weak alignment between institutions. Both situations require different recommendations. A constructive maturity framework must recognise these differences rather than reduce them to a single score.

2.10 Payer Interoperability

Health financing organisations are often treated as administrative actors. IML proposes a broader view. Public insurers, national health services, private insurers and mutual organisations are integral components of Health Information Ecosystems.

Payers possess information that frequently extends across time, providers and institutions. Reimbursement history, care utilisation, chronic disease pathways, medication coverage and long-term follow-up may reveal continuity patterns that no single healthcare provider can observe alone. When governed responsibly, this information can support prevention, care coordination, quality improvement and health system planning.

However, payer information is often separated from clinical information. Public and private payers may operate independently despite serving the same patient. Complementary insurers may hold information about services not visible to public systems. Public reimbursement databases may provide longitudinal data but lack clinical context. Clinicians may be unaware of administrative information that could support continuity, and payers may lack clinically meaningful feedback about outcomes.

Payer interoperability therefore does not mean that financial actors should access all clinical information. It means that appropriate, governed and purpose-limited information flows should exist where they improve care, prevention, safety or system performance. The principle remains the same: appropriate information, for legitimate purposes, under transparent governance.

IML proposes that payer interoperability be assessed through specific indicators: linkage between reimbursement and care pathways, coordination between public and private payers, availability of longitudinal utilisation information for authorised care coordination, contribution to prevention programmes, mechanisms for patient consent and transparency, and safeguards preventing misuse of health information for discriminatory purposes.

This dimension is particularly relevant for mixed health systems, but it also applies to predominantly public systems. Fragmentation may occur within public agencies, between regional structures, between insurance branches or between clinical and administrative databases. Payer interoperability is therefore not a private-sector issue. It is an ecosystem issue.

2.11 Artificial Intelligence as an Ecosystem Actor

Artificial intelligence is increasingly present in healthcare. It supports documentation, image interpretation, risk prediction, administrative workflows, research synthesis and decision support. Within IML, AI is not treated merely as a tool added to existing systems. It is considered an emerging actor within Health Information Ecosystems.

This does not mean that AI possesses clinical authority or ethical responsibility. Human professionals and institutions remain responsible for decisions. AI functions as an analytical and editorial instrument whose value depends entirely upon the quality, completeness, representativeness and governance of the information on which it operates.

Fragmented information weakens AI. Biased information distorts AI. Poorly governed information undermines trust. Conversely, mature Health Information Ecosystems can make AI safer and more useful by improving data quality, provenance, semantic consistency, clinical context and accountability.

The relationship between interoperability and AI is therefore reciprocal. Interoperability can improve the information environment in which AI operates. AI can help identify patterns, inconsistencies and opportunities for improvement within that environment. Both must remain governed by human responsibility, scientific evaluation and ethical oversight.

IML's approach to AI is inspired by a human-centred principle: advanced technologies should serve people and society. In this manuscript, the aspiration expressed in the OpenAI Charter that advanced artificial intelligence should benefit all of humanity is treated as an ethical inspiration, not as a claim of endorsement or partnership. The responsibility for IML's scientific content remains human.

2.12 Research Perspectives

Part II has introduced Health Information Ecosystems as the central unit of analysis for IML. This concept is both descriptive and programmatic. It describes the reality of modern healthcare, where information is produced and used by many actors. It also proposes a new way to assess maturity: not by examining isolated systems alone, but by evaluating the relationships, flows, governance and trust that connect them.

Several propositions emerge from this part. First, ecosystem maturity depends on relationships rather than organisations alone. Second, information flows should be assessed across their lifecycle: generation, validation, interpretation, sharing, reuse, preservation and transformation into knowledge. Third, clinical context preservation is essential for meaningful interoperability. Fourth, public and private payers should be recognised as integral ecosystem actors. Fifth, artificial intelligence should be evaluated according to the trustworthiness of the information environment in which it operates.

These propositions require further scientific validation. Future research should examine how Health Information Ecosystems can be measured, how institutional engagement can be operationalised, how payer interoperability can be assessed without compromising privacy, and how knowledge continuity can be linked to measurable health outcomes.

The following part will therefore move from ecosystem description to interoperability itself. If Part II defines the terrain, Part III defines the layers through which information must travel to become trustworthy knowledge.

Key Messages - Part II

- A Health Information Ecosystem is the appropriate unit of analysis for IML.
- The maturity of an ecosystem depends on relationships, not only on the quality of individual organisations or software applications.
- Information acquires value through circulation, interpretation, contextualisation and responsible reuse.
- Governance and trust are not secondary conditions; they are infrastructure.
- Institutional Engagement and Institutional Readiness should be treated as measurable dimensions of interoperability maturity.
- Payer Interoperability is a necessary but under-recognised component of health information ecosystems.
- Artificial intelligence depends on trustworthy, contextualised and governed information.
- Professional role alignment and workforce transformation should be treated as part of Health Information Ecosystem maturity.

PART III

Understanding Interoperability

Interoperability is not the movement of data alone. It is the capacity of an ecosystem to preserve meaning, trust and clinical usefulness across boundaries.

Part III begins the conceptual transition from ecosystem description to interoperability assessment. Part II defined the Health Information Ecosystem as the unit of analysis. Part III examines the capabilities that allow such an ecosystem to transform fragmented information into shared knowledge. Its purpose is not to replace existing interoperability standards, but to clarify what must be assessed when interoperability is understood as a condition for better health rather than as a technical feature.

3.1 The question interoperability should answer

Interoperability is often introduced through technical language: interfaces, standards, message formats, APIs, terminologies and data models. These elements are indispensable. However, they do not fully answer the central question of this manuscript.

For IML, the relevant question is not simply: can two systems exchange data?

The relevant question is: can this ecosystem preserve meaning, trust, clinical context and responsibility while information moves across organisational boundaries?

This distinction matters. A laboratory result may be transmitted successfully from one system to another and still fail to improve care if it arrives too late, lacks clinical context, uses ambiguous terminology, is not trusted by the receiving professional or cannot be acted upon within the workflow. Interoperability is therefore not merely the existence of exchange. It is the usefulness of exchange for health.

3.2 From exchange to meaningful use

The simplest form of interoperability enables transport. It allows information to move from one technical environment to another. This is necessary, but insufficient.

Meaningful interoperability requires that the receiving actor can understand the information, trust it, integrate it into a decision process and use it responsibly. It also requires that the rights, expectations and safety of the patient remain protected throughout the exchange.

IML therefore understands interoperability as a socio-technical capability. Technical exchange, semantic consistency, organisational coordination, institutional cooperation and clinical usefulness must work together. When one layer fails, the whole ecosystem may remain fragmented even if several technical standards are present.

3.3 The five layers of interoperability

IML proposes five complementary layers of interoperability: technical, semantic, organisational, institutional, and clinical/public health interoperability. These layers are not isolated levels arranged in a rigid hierarchy. They are interacting dimensions. Progress in one layer may be limited by weakness in another.

Technical interoperability concerns the ability of systems to exchange information through secure and reliable mechanisms. Semantic interoperability concerns the preservation of meaning through shared terminologies, data models and contextual interpretation. Organisational interoperability concerns workflows, roles and operational coordination. Institutional interoperability concerns governance, incentives, accountability and the willingness of institutions to collaborate. Clinical and public health interoperability concerns the capacity of exchanged information to support diagnosis, treatment, prevention, surveillance and learning.

3.4 Technical interoperability

Technical interoperability is the foundation upon which information exchange becomes possible. It includes secure connectivity, messaging, APIs, authentication, system availability, logging, performance and resilience. Without technical interoperability, information remains trapped inside local systems.

Technical interoperability is often the most visible dimension of digital health. It is also the dimension most easily confused with interoperability as a whole. A technically connected ecosystem may still fail to generate meaningful knowledge if semantic, organisational or institutional conditions are weak.

IML therefore evaluates technical interoperability as necessary but never sufficient. The question is not only whether systems can connect, but whether connections are reliable, secure, maintainable, documented and aligned with clinical use.

3.5 Semantic interoperability

Semantic interoperability concerns meaning. It ensures that information retains its interpretation when it moves between systems, organisations and professional communities.

A diagnosis, a laboratory value, a medication, an allergy or a microbiological result may appear simple. In practice, each may depend on terminology, units, reference ranges, coding systems, timestamps, specimen type, clinical indication and local conventions. Without semantic consistency, information can travel while meaning is lost.

For IML, semantic interoperability includes not only coded vocabularies and standard terminologies, but also the preservation of clinical context. In antimicrobial resistance, for example, a resistant isolate becomes clinically meaningful only when associated with symptoms, diagnostic interpretation, treatment and outcome. Semantic interoperability must therefore serve clinical reasoning, not merely data harmonisation.

3.6 Organisational interoperability

Organisational interoperability concerns the alignment of workflows, responsibilities and operational processes. Even when systems can exchange information and use compatible terminology, information may still fail to support care if organisations are not prepared to act upon it.

For example, a hospital discharge summary may be technically available to primary care, but if it arrives after the follow-up consultation, or if responsibility for medication reconciliation is unclear, the practical value of the information is reduced. Organisational interoperability therefore asks whether the right information reaches the right actor within the right workflow.

This layer is particularly important for continuity of care, chronic disease management, transitions between hospital and community care, laboratory reporting and public health surveillance.

3.7 Institutional interoperability

Institutional interoperability is one of the central proposals of IML. It refers to the capacity and willingness of institutions to cooperate, respond, share expertise, participate in governance and sustain collaboration over time.

Many ecosystems possess advanced technical capabilities but remain limited by institutional fragmentation. Organisations may have different mandates, legal interpretations, funding models, risk cultures or strategic priorities. These factors influence whether interoperability initiatives become practical reality.

Institutional interoperability should not be interpreted as a criticism of any specific country or organisation. It is a research hypothesis and a proposed assessment dimension. IML suggests that institutional engagement, responsiveness, transparency and participation in collaborative initiatives may be measurable determinants of interoperability maturity.

3.8 Clinical and public health interoperability

Clinical and public health interoperability represents the final test of ecosystem maturity. Information exchange is valuable only when it supports better care, prevention, surveillance, research or health system learning.

Clinical interoperability asks whether exchanged information improves diagnostic reasoning, treatment decisions, patient safety and continuity of care. Public health interoperability asks whether information can support surveillance, outbreak detection, policy evaluation and population-level prevention.

This layer connects the technical and institutional dimensions of interoperability to the core purpose of IML: protecting health. It prevents maturity assessment from becoming an inventory of technologies and redirects it towards measurable health usefulness.

3.9 The limits of interoperability

Interoperability does not mean that all information should circulate everywhere. Mature ecosystems are defined not only by their capacity to share, but also by their capacity to limit, protect and govern sharing.

Some information is sensitive. Some uses are inappropriate. Some actors do not need access. Some information should be anonymised, aggregated, delayed, restricted or not shared at all. The maturity of an ecosystem therefore depends upon its capacity to define legitimate purposes, authorisations, safeguards and accountability mechanisms.

This principle is essential for public trust. Interoperability without limits may become surveillance. Limits without interoperability may produce fragmentation. Mature ecosystems must find the ethical and operational balance between access and protection.

3.10 Transition to the IML Framework

Part III has reframed interoperability as a multidimensional ecosystem capability. Technical exchange remains essential, but it becomes only one component of a broader maturity model that also includes meaning, workflow, institutional engagement, clinical context, public health value and trust.

The following part will translate this understanding into the IML Framework. It will define the domains through which ecosystem maturity can be assessed, documented and progressively improved.

Layer	Core question
Clinical and public health	Does information improve care, prevention, surveillance or learning?
Institutional	Are institutions ready and willing to collaborate?
Organisational	Do workflows, roles and responsibilities support action?
Semantic	Is meaning preserved across systems and contexts?
Technical	Can systems exchange information securely and reliably?

Key Messages - Part III draft

- Interoperability is not only data exchange; it is the preservation of meaning, trust, context and usefulness across boundaries.
- Technical interoperability is necessary but insufficient.
- IML proposes five interacting layers: technical, semantic, organisational, institutional, and clinical/public health interoperability.
- Institutional interoperability is proposed as a measurable dimension of ecosystem maturity.
- Mature interoperability includes limits, authorisations and safeguards; unrestricted sharing is not the objective.

PART IV

The IML Framework

"A maturity framework should not decide who is best. It should help each ecosystem understand what to improve next."

4.1 Why a maturity framework?

The preceding parts established the conceptual foundations of IML. Health is the objective. Trustworthy information is the foundation. Interoperability is the path through which information can become shared knowledge and support better decisions.

A conceptual framework alone, however, is not sufficient. Health Information Ecosystems require instruments that make progress visible, comparable over time and actionable. A maturity framework provides such an instrument when it is designed not as a ranking device, but as a structured method for understanding strengths, weaknesses and realistic next steps.

IML therefore uses maturity assessment as a means of continuous improvement. Its purpose is to help ecosystems identify where information loses meaning, where governance is incomplete, where trust is fragile, where technical exchange is insufficient and where institutional collaboration remains underdeveloped. The maturity score is not the objective. The improvement pathway is.

4.2 From ranking to continuous improvement

Many maturity models create an implicit hierarchy. Countries, organisations or programmes are compared according to scores that may be attractive for communication but insufficient for transformation. Such comparisons can be useful when they stimulate learning, but they become counterproductive when they obscure context.

IML adopts a different philosophy. Its purpose is not to classify countries as successful or unsuccessful. Every ecosystem is historically, legally, economically and institutionally specific. A high-income country may possess sophisticated infrastructure while remaining fragmented. A lower-resource country may lack advanced systems while demonstrating strong institutional alignment and pragmatic interoperability.

For this reason, IML assessment should always lead to three practical questions: Where are we today? What should improve next? How can progress be measured over time? This approach transforms maturity assessment into a cycle of learning rather than a competition.

4.3 The six IML domains

The IML Framework is organised around six core domains. These domains provide the main structure for assessing the maturity of Health Information Ecosystems.

Domain 1: Governance and Standards.
Domain 2: Technical Interoperability.
Domain 3: Identity, Consent and Trust.
Domain 4: Adoption and Use.
Domain 5: Security and Resilience.
Domain 6: Feedback, Correction and Learning.

These domains should not be interpreted as isolated compartments. They interact continuously. Governance influences technical choices. Identity and consent influence trust. Adoption determines whether standards become useful in practice. Security protects continuity. Feedback and correction allow ecosystems to learn. Together, the six domains form a practical architecture for improvement.

4.4 Domain 1 - Governance and Standards

Governance determines how decisions are made, who is responsible, how conflicts are resolved and how standards are selected, maintained and implemented. Without governance, interoperability may remain a technical aspiration without institutional authority.

This domain examines the existence of national or regional strategies, standards governance, terminology policies, implementation guides, public accountability mechanisms and transparent decision-making processes. It also considers whether governance includes multiple stakeholders: clinicians, patients, laboratories, payers, researchers, software developers and public authorities.

A mature ecosystem does not merely publish standards. It defines how standards are chosen, updated, implemented, monitored and corrected. Governance gives interoperability a memory, a direction and a method of accountability.

4.5 Domain 2 - Technical Interoperability

Technical interoperability concerns the capacity of systems to exchange information through reliable, documented and secure mechanisms. It includes APIs, messaging standards, data transport, identifiers, infrastructure, network availability and system integration.

This domain does not evaluate technology for its own sake. A technically sophisticated system may still fail if exchanged information is incomplete, poorly governed or clinically meaningless. Conversely, simple technical solutions may create significant value when they support real clinical workflows and are embedded in strong governance.

IML therefore evaluates technical interoperability according to its contribution to meaningful information movement across the ecosystem.

4.6 Domain 3 - Identity, Consent and Trust

Health information cannot circulate responsibly without reliable mechanisms for identity, consent, authorisation and trust. Identity allows information to be associated with the correct person, organisation or professional. Consent and authorisation define the conditions under which information may be accessed, shared or reused. Trust determines whether patients, professionals and institutions accept the legitimacy of these mechanisms.

This domain includes patient identity, professional identity, organisational identity, consent management, access rights, auditability and transparency. It also includes the capacity of the ecosystem to explain to patients how their information is used, protected, corrected and governed.

Digital identity initiatives are essential, but IML emphasises that identity alone is insufficient. Identity must be embedded within governance, ethics, legal safeguards, institutional responsiveness and practical mechanisms for correction. For this reason, IML should not be understood as proposing a new personal identity number, a replacement for national identity systems or a centralised health identity registry.

A more appropriate research direction is the exploration of a secure identity-access layer capable of working with existing national identifiers and emerging digital identity infrastructures. In this model, national identifiers remain under the authority of the relevant national systems. IML would not store, expose or replace the national identifier itself. Instead, an authorised application or trusted service could transform a verified identity assertion into a purpose-limited, auditable and privacy-preserving access token for a specific health information use.

Where IML materials discuss a federated interoperability identifier, that proposal should be read as an illustrative research placeholder for linkage, deduplication and trust mechanisms, not as an operational proposal for a new personal identity number. The preferred framing is therefore not a new identifier, but a secure access layer able to use existing identity systems without unnecessarily exposing sensitive identity data.

Such uses could include care coordination, AMR/BMR case review, public health reporting, cross-system deduplication, research linkage under governance or patient-mediated access to health information. The token would not be a new identity number. It would be a contextual proof that an identity or authorisation has been verified for a defined purpose, under defined rules and for a defined duration.

IML should distinguish four elements that are often confused: identity, identifier, access token and carrier mechanism. Identity refers to the verified person, professional or organisation. An identifier refers to the number or attribute used by a national, local or organisational system. An access token is a temporary, signed and auditable proof that access has been authorised for a defined purpose. A carrier mechanism, such as a QR code or mobile application, merely transports or presents that token. None of these elements should be treated as equivalent.

Several identity-light mechanisms may be explored without creating a new universal identifier. These include temporary access tokens, patient-mediated authorisation through a mobile application or digital wallet, trusted identity brokers, contextual pseudonyms, project-specific hashes with appropriate governance, and episode-based linkage for selected use cases. The common objective is to expose the least identity information necessary while still allowing trustworthy linkage, accountability and correction.

A QR code could serve as a practical carrier for a temporary verification token, but it should not contain sensitive identity or health information in clear text. The QR code should be understood as an access mechanism, not as the identity itself. Its content should be temporary, signed, auditable and revocable. A QR code that exposes a national identifier or medical information in clear text would contradict the IML principles of proportionality, privacy and trust.

Biometric authentication, where used, should remain local to the user's device and serve only to unlock the application or confirm user presence. It should not become a central biometric health identifier. This distinction is essential for avoiding excessive surveillance, re-identification risks and exclusion of individuals unable or unwilling to use biometric mechanisms.

This approach is compatible with the broader European movement toward trusted digital identity and health data governance. The European Digital Identity Wallet is intended to provide a safe, reliable and private means of digital identification, with users able to prove who they are and share selected identity attributes across digital services. The European Health Data Space aims to establish a common framework for the use and exchange of electronic health data across the European Union, strengthening individuals' access to and control over their health data while enabling certain forms of reuse for public interest, policy support and scientific research. IML would not replace these frameworks. It would offer a research pathway for studying how identity, consent, access, auditability and health information continuity interact in concrete use cases.

Before any operational implementation, such an identity-access layer would require scientific validation, privacy and security assessment, transparent governance, correction procedures, safeguards against exclusion or misuse, legal review within each relevant jurisdiction, and alignment with European and national identity and health data frameworks.

At this stage, IML does not propose an operational identity system. It proposes a research hypothesis for discussing trust, pseudonymisation, deduplication, auditability, patient-mediated access and cross-system linkage within Health Information Ecosystems.

This research direction should be read in relation to emerging European digital identity and health data governance frameworks, especially the European Digital Identity Wallet and the European Health Data Space.

4.7 Domain 4 - Adoption and Use

Interoperability is only meaningful when it is used. Many systems are technically capable of exchanging information but remain underused because workflows, incentives, training or professional trust are insufficient.

This domain evaluates whether interoperable services are adopted by clinicians, patients, laboratories, payers and public health authorities. It examines usability, workflow integration, training, incentives, professional acceptance and perceived usefulness.

A mature ecosystem reduces the burden on professionals rather than adding new administrative complexity. Adoption should therefore be assessed not merely by connection rates, but by whether information exchange improves decisions, saves time and supports safer care.

Adoption should also include professional role alignment. Digital tools are not adopted in a vacuum. They are used by professions whose training, legal responsibilities, access rights, remuneration and symbolic status may not correspond to the real distribution of work in modern healthcare. IML should therefore assess whether doctors, nurses, pharmacists, laboratory professionals, allied health professionals, public health workers, data professionals and administrative actors are trained and authorised to contribute appropriately to information quality, correction, interpretation and reuse.

4.8 Domain 5 - Security and Resilience

Health Information Ecosystems require security because health information is sensitive. They also require resilience because health services must continue during disruptions, cyberattacks, disasters or institutional failures.

This domain examines cybersecurity, continuity planning, incident response, data backup, service availability, audit trails and recovery mechanisms. It also considers whether security measures preserve clinical usefulness. Excessive restrictions may protect information while preventing appropriate care; insufficient restrictions may enable misuse.

IML therefore treats security as part of trust. Security should enable responsible information use, not merely prevent exchange.

4.9 Domain 6 - Feedback, Correction and Learning

No Health Information Ecosystem is perfect. Errors occur. Data become outdated. Standards evolve. Clinical knowledge changes. A mature ecosystem must therefore be able to detect, correct and learn from its own limitations.

This domain evaluates mechanisms for data correction, patient feedback, professional feedback, incident reporting, quality monitoring, outcome measurement and continuous improvement. It also includes the capacity to revise indicators, update implementation guides and document lessons learned.

Feedback transforms interoperability from a static infrastructure into a learning system. Without correction, errors circulate. With correction, information becomes more trustworthy over time.

4.10 Cross-cutting dimension - Institutional Engagement

The six domains describe core components of maturity, but IML also recognises cross-cutting dimensions that influence all domains. Institutional Engagement is the first of these dimensions.

Institutional Engagement refers to the willingness and capacity of organisations to participate in dialogue, share expertise, respond to collaboration requests, contribute to shared governance and engage in practical implementation. Technical readiness without institutional engagement may produce formal interoperability with limited real-world effect.

IML proposes that Institutional Engagement should be studied as an independent determinant of interoperability maturity. It should not be used to criticise specific institutions. Rather, it should help understand why some ecosystems transform available standards into practice while others remain fragmented despite technical resources.

4.11 Cross-cutting dimension - Payer Interoperability

Public insurers, national health services, private insurers and mutual organisations are not peripheral actors. They possess longitudinal information about care pathways, reimbursement, chronic disease management, prevention programmes and health system utilisation.

Payer interoperability therefore deserves explicit attention. In many ecosystems, clinical data and financing data remain separated even when both concern the same patient journey. This separation may limit prevention, continuity of care, evaluation of outcomes and understanding of system performance.

IML does not propose unrestricted exchange between payers and clinical care. On the contrary, it emphasises governance, proportionality, patient rights and purpose limitation. The question is not whether all payer information should circulate. The question is which information, under which conditions, can responsibly support better health.

4.12 Cross-cutting dimension - AI Readiness

Artificial intelligence increasingly depends on the quality of Health Information Ecosystems. AI systems trained or deployed on fragmented, biased or poorly contextualised information may amplify existing weaknesses. Conversely, trustworthy, interoperable and well-governed information can support safer and more useful AI.

AI Readiness therefore should not be limited to the presence of algorithms. It should evaluate data quality, context preservation, governance, transparency, human oversight, auditability, bias monitoring and clinical responsibility.

Within IML, AI is considered an ecosystem actor and an analytical instrument. It does not replace scientific judgment, clinical responsibility or ethical governance.

4.13 The IML maturity logic

The IML Framework does not assume a single path to maturity. Different ecosystems may progress in different sequences. Some may begin with governance. Others may begin with basic technical exchange. Others may prioritise identity, cybersecurity or specific clinical use cases.

The maturity logic is therefore progressive. Each assessment should identify the current profile of the ecosystem, the most relevant next step and the evidence required to measure progress. This approach allows IML to support both low-resource settings and highly digitalised but fragmented systems.

Maturity should be understood as the capacity to learn, adapt and improve, not as the possession of a fixed technological architecture.

4.14 Transition to the methodology

Part IV has introduced the conceptual structure of the IML Framework. The next part translates this structure into methodology. It will address indicators, evidence grading, data sources, scoring philosophy, weighting, validation and the role of expert review.

The transition from framework to methodology is essential. Without methodology, concepts remain inspirational. With methodology, they become discussable, testable and improvable.

The purpose of the next part is therefore to transform the IML vision into an assessment process capable of supporting continuous improvement.

A six-domain hexagon should be created for this section. The central concept should be Health Information Ecosystem. Around it: Governance and Standards; Technical Interoperability; Identity, Consent and Trust;

Adoption and Use; Security and Resilience; Feedback, Correction and Learning. Three cross-cutting bands should surround the hexagon: Institutional Engagement, Payer Interoperability and AI Readiness.

Key Messages - Part IV

- The IML Framework is designed for continuous improvement rather than country ranking.
- The six IML domains provide a practical structure for assessing Health Information Ecosystem maturity.
- Governance, standards, technology, identity, adoption, security and feedback must be evaluated together.
- Institutional Engagement, Payer Interoperability and AI Readiness are cross-cutting dimensions that influence all domains.
- Maturity should be understood as the capacity of an ecosystem to learn, adapt and improve over time.
- Adoption and use require alignment between digital tools and the real distribution of professional roles, responsibilities, access rights and training.

PART V

Methodology

A maturity framework is useful only if it helps institutions understand where they are, what should improve next and how progress can be demonstrated over time.

5.1 Why methodology matters

The previous parts established the conceptual foundations of IML. Health comes before technology. Health information should be understood as part of an ecosystem. Interoperability should be assessed not only as a technical capacity, but as a practical condition for transforming fragmented information into trusted knowledge that can protect health.

These principles require a methodology. Without methodology, IML would remain a vision. With methodology, it becomes a tool for structured assessment, discussion, comparison over time and continuous improvement.

The purpose of the IML methodology is not to produce a political ranking of countries or institutions. Its purpose is to support learning. A maturity assessment should help an ecosystem understand its strengths, identify its bottlenecks and define realistic next steps. For this reason, IML treats measurement as the beginning of improvement, not as its conclusion.

5.2 From concepts to indicators

Every maturity framework depends on the quality of its indicators. Indicators should not merely count the existence of digital tools. They should assess whether the ecosystem is capable of producing, validating, exchanging, interpreting and reusing trustworthy information in ways that improve health.

IML therefore proposes that each indicator should be designed according to five criteria.

1. Relevance: the indicator should be clearly linked to the protection or improvement of health.
2. Measurability: the indicator should be assessable through documented evidence rather than general impressions.
3. Comparability: the indicator should allow comparison over time and across contexts without ignoring national diversity.
4. Actionability: the indicator should point towards a possible improvement pathway.
5. Transparency: the rationale, evidence and scoring logic should be understandable to external reviewers.

The indicator is therefore not a bureaucratic unit. It is a bridge between a concept and an action. If an indicator cannot guide improvement, it should not be retained.

5.3 Evidence before opinion

IML should avoid relying exclusively on self-declaration. Expert opinion is valuable, especially when evaluating institutional engagement, clinical workflows or emerging practices, but it should be distinguished from documented evidence.

For this reason, each assessment should record the type of evidence supporting every indicator. Possible evidence categories include policy documents, legislation, technical documentation, implementation reports, scientific publications, audit results, interviews, public datasets and direct demonstration of operational capability.

The same score may have different credibility depending on the strength of the supporting evidence. A documented national implementation should not be treated in the same way as an aspirational policy statement. Conversely, the absence of published documentation does not necessarily mean the absence of capability. It means that the level of evidence is uncertain and should be explicitly recorded.

5.4 Proposed evidence grading

IML proposes a simple evidence grading system for early versions of the framework. This system can be refined during external review.

Grade	Evidence level	Interpretation
E0	No evidence identified	No public or provided evidence supports the indicator.
E1	Statement or intention	A policy statement, strategy or declaration exists, but implementation is not documented.
E2	Partial implementation evidence	Pilot projects, local implementation or limited operational evidence exist.
E3	Documented operational implementation	The capability is implemented and documented in routine practice.
E4	Evaluated implementation	The implementation has been evaluated, audited or described in scientific or institutional reporting.
E5	Mature and reusable evidence	Evidence demonstrates sustained use, governance, monitoring and capacity for reuse or replication.

This grading system should not be confused with scientific certainty. It is a practical method for distinguishing between absence of evidence, evidence of intention, evidence of implementation and evidence of mature use.

5.5 Scoring philosophy

IML scoring should reflect progressive maturity rather than binary compliance. An ecosystem should not be described as simply interoperable or non-interoperable. Most systems are partially mature. They may be advanced in technical infrastructure while weaker in institutional engagement. They may possess strong governance but limited semantic interoperability. They may support hospital exchange while neglecting primary care, public health or payer interoperability.

For this reason, IML proposes a maturity scale that recognises progression.

Maturity level	Name	Description
Level 0	Absent or undocumented	No meaningful evidence of capability.
Level 1	Initial	The capability exists in intention, planning or isolated form.
Level 2	Emerging	Pilot implementations or limited operational use exist.
Level 3	Operational	The capability is routinely used in defined settings.
Level 4	Integrated	The capability is coordinated across organisations or sectors.
Level 5	Learning ecosystem	The capability supports monitoring, feedback, reuse and continuous improvement.

The highest level is deliberately not named perfect. It is named learning ecosystem. This reflects the IML philosophy: maturity is not the end of change, but the capacity to keep improving.

5.6 Country profiles rather than country rankings

Country rankings can attract attention, but they often simplify complex realities. IML should avoid reducing health information ecosystems to a single competitive score. A country may be strong in microbiology surveillance but weak in payer interoperability. Another may have advanced digital identity but limited clinical context preservation. A third may have modest resources but strong institutional engagement and rapid learning capacity.

IML therefore proposes country profiles rather than simple rankings. A profile describes maturity across multiple domains and identifies priority pathways for improvement. This approach is more respectful of national diversity and more useful for policy action.

The purpose is not to decide which country is best. The purpose is to understand what each ecosystem can improve next.

5.6.1 Complementarity with the Global Digital Health Monitor and maturity dashboards

IML country profiles should not duplicate the Global Digital Health Monitor (GDHM) or other high-level digital health maturity dashboards. Their purpose is not to replace instruments that monitor national digital health capacity, but to add a clinically oriented and ecosystem-oriented layer of interpretation. Where existing tools describe whether digital health capacity exists, IML examines how that capacity performs in concrete information pathways, such as the AMR/BMR demonstrator.

This distinction should remain explicit in every IML assessment: existing maturity dashboards describe digital health capacity, while IML examines interoperability usefulness, clinical context preservation, feedback, correction, payer interoperability and learning in concrete health information pathways.

5.6.2 Comparative positioning: GDHM and IML

A reviewer may ask why IML should not simply copy the Global Digital Health Monitor. The answer is that GDHM and IML do not answer the same question. GDHM monitors the enabling environment and maturity of digital health at country, regional and global levels. IML examines whether that capacity preserves clinical meaning, context, trust, auditability and usefulness in concrete health information pathways.

Comparison point	Global Digital Health Monitor (GDHM)	Interoperability Maturity Lab (IML)
Primary purpose	Monitors digital health maturity and the enabling environment at country, regional and global levels.	Assesses whether health information ecosystems transform fragmented information into trustworthy, contextualised and actionable knowledge.
Unit of analysis	Countries, regions and year-on-year progress.	Health Information Ecosystems, actors, relationships, information flows and clinical or public-health pathways.
Core question	How mature is the digital health enabling environment?	Does digital health capacity preserve meaning, context, trust and usefulness across boundaries?
Indicator logic	Standardised indicators for governance, strategy, policy, workforce, standards and interoperability, infrastructure, services and applications, with cross-cutting issues such as equity and emerging technology.	Domains and cross-cutting dimensions focused on interoperability usefulness, clinical context, correction, institutional engagement, payer interoperability, AI readiness and technical quality.
Role of interoperability	One component of a broader digital health maturity landscape.	The central object of analysis: preservation of meaning, context, responsibility and usefulness.
Clinical test	Clinical usefulness may be implied, but it is not the main level of analysis.	Use cases such as UTI / MDR E. coli test whether the pathway remains coherent from symptoms to surveillance and learning.
Expected output	Dashboards, visualisations and country-level comparisons.	Maturity profiles, bottleneck analysis, improvement pathways, demonstrators and implementation guidance.
Why not copy GDHM?	GDHM already provides high-level monitoring of	Copying GDHM would duplicate an established

Comparison point	Global Digital Health Monitor (GDHM)	Interoperability Maturity Lab (IML)
	national digital health capacity.	instrument. IML adds a complementary clinical and ecosystem layer focused on interoperability usefulness.
Validation status and evidence transparency	Country information depends on survey submission and official validation processes. The WHO beta platform notes that data are pending review and validation by participating Member States before replacing the existing platform; during this review, visibly government-approved country-level validation appeared limited, with Austria identified as the only clearly government-validated country entry reviewed.	Uses evidence grading to distinguish official validation, documented implementation, expert-informed evidence, exploratory evidence and research hypotheses. IML can go further where dashboard data are incomplete, pending validation or too high-level for clinical interoperability analysis.

This distinction should be maintained in IML communications: existing maturity dashboards describe digital health capacity; IML examines interoperability usefulness in health information ecosystems.

5.6.3 Validation status of GDHM data and the added role of IML

The WHO-hosted GDHM beta explicitly signals that data require review and validation by participating Member States before replacement of the existing platform. In the materials reviewed for this manuscript version, government-approved country-level validation appeared limited, with Austria identified as the only clearly government-validated country entry. This observation should be updated as the GDHM validation process evolves.

A reviewer may therefore ask whether IML should simply copy GDHM once more countries are validated. The answer is no. GDHM validation strengthens the global maturity map, but it does not by itself answer whether interoperability preserves clinical context, links laboratory information to diagnosis and treatment, supports correction, or transforms information into usable knowledge in a defined care pathway. IML is designed to examine these deeper questions.

This is the practical niche for IML. Instead of waiting for every national dataset to be officially validated, IML can work with clearly graded evidence: documented facts, government-validated information, expert-informed observations, public documentation, implementation reports and research hypotheses. The result is not a substitute for official validation, but a transparent analytical layer that can reveal what remains unknown and what should be examined next.

5.7 Weighting and transparency

A maturity framework must decide whether all indicators have equal weight or whether some should contribute more strongly to the overall interpretation. IML should treat weighting cautiously. Premature weighting can create an illusion of mathematical precision. Equal weighting, however, may also be misleading if some dimensions are more important for patient safety, public health or clinical usefulness.

In early versions, IML may use transparent unweighted domain scores while testing alternative weighting models. Future weighting should be justified by expert consultation, empirical validation, clinical relevance and sensitivity analysis. No weighting model should be hidden. Readers should be able to understand how any score was produced.

> *In IML, transparency is more important than numerical elegance.*

5.8 Data sources

IML assessments should combine multiple sources. No single source can adequately describe a Health Information Ecosystem. Public documentation may reveal policy and governance. Technical specifications may reveal interoperability capacity. Scientific publications may describe implemented infrastructures. Interviews may reveal institutional readiness. Demonstrations may confirm operational capability. Routine data may show actual use.

A mature assessment should therefore triangulate evidence rather than rely on a single channel. When evidence is missing, uncertain or inaccessible, the assessment should say so explicitly. Lack of transparency is itself relevant to maturity, but it should be interpreted carefully and without assuming institutional failure.

5.9 Expert review

Because health information ecosystems are socio-technical systems, expert review is essential. Technical experts may evaluate standards and architecture. Clinicians may assess usefulness and workflow integration. Public health specialists may assess surveillance value. Legal experts may evaluate governance. Patients and civil society may evaluate trust, consent and equity.

IML should therefore encourage multidisciplinary review. A methodology that excludes clinicians risks becoming technically elegant but clinically weak. A methodology that excludes technical expertise risks becoming aspirational. A methodology that excludes patients risks losing legitimacy.

5.10 Validation strategy

The IML methodology should be validated progressively. Initial validation may begin with expert review and comparison with existing frameworks. Subsequent validation should test whether IML assessments identify real bottlenecks and whether recommended actions lead to measurable improvement.

The antimicrobial resistance demonstrator provides an appropriate first validation pathway. AMR requires integration of microbiology, clinical context, treatment, outcomes and public health surveillance. If IML can help clarify the difference between resistant isolates and clinically meaningful infections, it will demonstrate practical value beyond abstract assessment.

5.11 Versioning and reproducibility

The IML framework should evolve through versioned releases. Each version should document changes in definitions, indicators, scoring logic and evidence requirements. This is essential for reproducibility. If a country assessment changes over time, reviewers should be able to determine whether the change reflects real progress or a modification of the methodology.

Version control also supports scientific humility. IML should not claim to be complete at its first release. It should be designed to learn from critique, implementation and empirical testing.

5.12 Methodological limitations

Every maturity framework has limitations. Some indicators may be easier to measure in high-resource countries than in low-resource settings. Publicly available information may be incomplete. Institutional engagement may be difficult to quantify. Legal frameworks may differ substantially across countries. Implementation may exist locally without national documentation. Conversely, national strategies may exist without meaningful operational use.

IML should make these limitations visible rather than conceal them. A transparent limitation strengthens a methodology more than an unjustified claim of certainty.

5.13 Transition to the demonstrator

The methodology described in this part provides the bridge between concept and implementation. It transforms the IML vision into a structured assessment process. The next step is to test this process against a concrete health problem where interoperability matters directly.

Antimicrobial resistance provides that test. It is global, clinically meaningful, scientifically documented and highly dependent on the integration of information across laboratories, clinicians, public health authorities and health systems. For this reason, the next part applies the IML perspective to AMR and multidrug-resistant bacteria as the first demonstrator of the framework.

Measure -> Understand -> Recommend -> Implement -> Evaluate -> Learn

Key Messages - Part V

- The purpose of IML methodology is continuous improvement, not ranking.
- Indicators should be relevant, measurable, comparable, actionable and transparent.
- Evidence grading should distinguish intention, implementation and evaluated routine use.
- Country profiles are more useful than league tables.
- Weighting should remain transparent and subject to validation.
- Methodological humility is essential for scientific credibility.
- AMR provides the first practical validation pathway for the IML framework.

PART VI

Demonstrator: Antimicrobial Resistance and Multidrug-Resistant Bacteria

A scientific framework becomes stronger when it is tested against a problem where clinical care, microbiology, public health, governance and information systems all meet.

6.1 Why a demonstrator matters

Every scientific and operational framework needs a thread. Without a concrete case study, even the most coherent conceptual model risks remaining abstract. A demonstrator provides a common object around which definitions, indicators, data flows, governance questions and implementation choices can be tested.

IML therefore considers that its first practical development should be anchored in a clinical and public health use case. The purpose of such a use case is not to reduce the scope of IML. It is to give the framework an initial point of departure, a shared reference and a way to evaluate whether the proposed concepts can actually help solve a recognised problem.

A case study should fulfil several conditions. It should address an important health issue. It should require several actors to collaborate. It should depend on both clinical and non-clinical information. It should reveal the limits of fragmented data. It should also be sufficiently concrete to support pilot implementation. Antimicrobial resistance, and particularly multidrug-resistant bacteria, fulfils these conditions with unusual strength.

For this reason, IML proposes antimicrobial resistance and multidrug-resistant bacteria as the first demonstrator of the framework. In French-speaking contexts, the expression BMR is commonly used for bacteries multiresistantes. In this manuscript, AMR and MDRO/BMR will be used together when useful, because the clinical problem is both global and immediately recognisable to clinicians.

6.2 Why antimicrobial resistance is an appropriate starting point

Antimicrobial resistance is not only a microbiological phenomenon. It is a clinical, epidemiological, organisational and societal challenge. It affects individual patients, healthcare teams, laboratories, hospitals, primary care, pharmacies, infection control services, public health agencies, payers and policymakers.

This multidimensional character makes AMR particularly suitable for IML. A resistant isolate may be identified in a laboratory, but its meaning depends on clinical context. Is the patient symptomatic? Is the result compatible with infection, colonisation, contamination or asymptomatic bacteriuria? Was an antibiotic prescribed? Was the treatment appropriate? Did the patient improve? Was there a recurrence? Did the case represent a public health signal?

None of these questions can be answered by microbiology alone. Conversely, clinical care without microbiological information may lead to inappropriate treatment. AMR therefore exposes the limits of fragmented information in a particularly visible way. This is precisely why it can serve as a starting point. It requires the ecosystem to work.

6.3 From resistant isolate to clinically meaningful infection

A central difficulty in AMR surveillance is the distinction between microbiological detection and clinically meaningful disease. A resistant bacterium identified in a sample does not automatically represent symptomatic infection. It may reflect colonisation. It may reflect contamination. In urinary samples, it may represent asymptomatic bacteriuria rather than a urinary tract infection requiring treatment.

This distinction is not academic. It has direct consequences for clinical decision-making, antibiotic stewardship, surveillance, research and policy. Counting resistant isolates is not equivalent to measuring the incidence of infections caused by resistant organisms.

IML therefore proposes that the first AMR demonstrator should focus on the transformation of isolated laboratory data into clinically meaningful information. This requires connecting microbiological results with symptoms, diagnosis, treatment and outcomes.

The goal is not simply to ask whether a bacterium is resistant. The goal is to ask whether the patient has a clinically relevant infection caused by a resistant organism, whether the treatment was appropriate, and whether the outcome was favourable.

6.4 The urinary tract infection use case

Urinary tract infections provide a strong initial use case for several reasons. They are frequent. They occur in primary care, emergency departments, hospitals, nursing homes and community settings. They involve laboratory testing, clinical interpretation, prescribing decisions and recurrent care pathways. They also raise persistent questions regarding asymptomatic bacteriuria, overdiagnosis and antimicrobial stewardship.

Escherichia coli is particularly relevant as an initial focus because it is one of the most common bacterial causes of urinary tract infection and frequently appears in antimicrobial resistance surveillance. Multidrug-resistant E. coli urinary isolates therefore create a practical setting in which IML can test its core hypothesis: meaningful health information requires the preservation of clinical context.

The urinary tract infection use case is also understandable to clinicians, laboratories, public health authorities and payers. It is neither rare nor exotic. It belongs to everyday medicine. This ordinariness is an advantage. If interoperability cannot improve the management and measurement of a common condition, it is unlikely to transform more complex areas of care.

6.5 Minimum clinical dataset for the AMR demonstrator

A practical demonstrator requires a minimum dataset. The objective is not to collect every possible variable, but to identify the smallest set of information required to distinguish clinically meaningful infection from other situations.

For multidrug-resistant E. coli urinary tract infections, IML proposes the following minimum clinical dataset as a starting point for discussion and validation.

6.6 Proposed minimum dataset elements

The proposed dataset should include symptoms, fever, urinary signs, context of complicated infection, pregnancy when relevant, catheter use when relevant, immunosuppression when relevant, recent hospitalisation, leucocyturia, culture result, bacterial count, species identification, antimicrobial susceptibility testing, clinical diagnosis retained by the clinician, treatment prescribed, treatment modification after results, clinical evolution and recurrence when available.

This dataset should not be interpreted as a final standard. It is an IML proposal designed to stimulate discussion. Its purpose is to demonstrate that the value of microbiology increases when it is connected to clinical information.

6.7 Microbiology, terminology and semantic consistency

A demonstrator cannot rely only on free text. Microbiology data require consistent terminology for organism identification, sample type, bacterial count, resistance mechanisms, susceptibility results and interpretation categories. Clinical data require comparable consistency for symptoms, diagnosis, treatment and outcomes.

The AMR demonstrator should therefore become a test bed for semantic interoperability. It should examine whether existing terminologies and coding systems can adequately represent the information required for clinically meaningful AMR measurement.

A key question is whether the ecosystem can preserve meaning across systems. A urine culture result entered in a laboratory system should remain interpretable when reused by a general practitioner, an infection control team, a public health agency, a payer or a research group.

6.8 Information flows in the AMR demonstrator

The AMR demonstrator should map information flows rather than databases alone. A patient presents with symptoms. A clinician orders a urine test. A sample is collected. The laboratory produces a result. The clinician interprets the result. Treatment may be started, changed or stopped. The information may be reused for surveillance, stewardship, reimbursement, research and quality improvement.

Each step represents a potential rupture. The clinical indication may not reach the laboratory. The laboratory result may not be available in time. The diagnosis may not be linked to the culture. The prescription may not be connected to susceptibility data. The outcome may disappear from the record. Public health surveillance may receive the isolate but lose the clinical story.

The demonstrator should therefore ask not only whether data exist, but whether the complete informational pathway remains coherent.

6.9 Payers and AMR

Public and private payers are often overlooked in AMR interoperability. Yet they may contribute important longitudinal information regarding consultations, antibiotic prescriptions, hospitalisations, repeated testing and care pathways. In some systems, reimbursement data may help identify repeated antibiotic exposure, recurrence, transitions between providers or fragmented follow-up.

The inclusion of payers does not mean that reimbursement data should replace clinical data. It means that mature Health Information Ecosystems should define how payer information can support prevention, stewardship and continuity of care under appropriate governance.

This is particularly relevant for AMR because resistant infections often reflect cumulative exposure, repeated care contacts and system-level patterns. Payer interoperability may therefore contribute to understanding the pathway that precedes infection, not only the episode itself.

6.10 Artificial intelligence and AMR

Artificial intelligence may support AMR work through pattern recognition, risk prediction, decision support, surveillance and research. However, AI cannot compensate for poor data quality, missing clinical context or weak governance. An algorithm trained on isolated microbiology results may reproduce the same limitations as the fragmented systems that generated them.

The AMR demonstrator should therefore treat AI readiness as dependent on ecosystem maturity. Before asking whether AI can predict resistant infection, the ecosystem must ask whether the information used by AI is accurate, contextualised, timely, representative and ethically governed.

In this sense, AMR provides a useful test of responsible AI. It shows that artificial intelligence in health is only as reliable as the information ecosystem on which it depends.

6.11 What the AMR demonstrator should measure

The AMR demonstrator should not only measure the presence of resistant organisms. It should measure the capacity of an ecosystem to transform fragmented information into clinically meaningful knowledge.

Possible indicators include the proportion of microbiology results linked to a clinical indication, the availability of symptoms at the time of interpretation, the link between culture and diagnosis, the link between susceptibility testing and treatment, the delay between sample collection and clinical action, the availability of outcome information, the distinction between infection and colonisation, and the ability to generate surveillance indicators that reflect symptomatic infection rather than isolates alone.

These indicators should be refined through expert review. Their value lies not in producing a perfect score, but in identifying where the ecosystem loses meaning.

6.12 Why BMR can become the starting point of IML activity

A framework becomes credible when it demonstrates practical usefulness. BMR offers such an opportunity. It is important enough to justify attention, concrete enough to support implementation, and complex enough to test the full IML logic.

It connects the central concepts of the manuscript: Health Information Ecosystems, Continuity of Health Information, Clinical Context Preservation, Knowledge Continuity, Institutional Engagement, Payer Interoperability and responsible AI.

It also allows IML to begin with a service that has immediate practical relevance. A project that helps clinicians, microbiologists and public health teams better distinguish colonisation from infection, resistant isolate from symptomatic disease, and data accumulation from meaningful knowledge would already provide value.

For this reason, BMR should not be presented as an isolated example. It should be presented as the first operational thread of IML. A thread that connects theory to practice. A thread that allows the framework to be tested, criticised, improved and eventually extended to other domains.

6.13 Transition to implementation

The AMR demonstrator marks the transition from conceptual framework to implementation. It shows that IML is not intended to remain a theoretical model. Its purpose is to guide practical improvement within real Health Information Ecosystems.

The following part therefore moves from demonstration to action. It examines how assessment can lead to recommendations, how recommendations can lead to pilots, and how pilots can prepare the development of an Open Clinical Workspace capable of supporting interoperability without replacing existing systems.

Table 6.1 Proposed minimum dataset for MDR E. coli urinary tract infection

Category	Data elements	Purpose
Clinical presentation	Symptoms, urinary signs, fever, duration, severity	Distinguish symptomatic infection from colonisation or asymptomatic bacteriuria
Risk and context	Catheter, pregnancy, immunosuppression, recent hospitalisation, recurrent UTI, complicated context	Interpret risk and clinical significance
Laboratory data	Sample type, leucocyturia, culture result, bacterial count, species, susceptibility testing	Connect microbiology with clinical interpretation
Clinical decision	Diagnosis retained, treatment prescribed, treatment modified after result	Evaluate decision-making and antimicrobial stewardship
Outcome	Clinical improvement, recurrence, hospitalisation, adverse events when available	Measure meaningful health impact
System information	Provider setting, delays, data availability, linkages between systems	Assess ecosystem performance

Key Messages - Part VI

- A maturity framework needs a concrete thread that can test its concepts against real-world complexity.
- AMR and BMR provide a strong starting point because they connect clinical care, microbiology, public health, payers, governance and information systems.
- A resistant isolate is not equivalent to a clinically meaningful infection.
- The AMR demonstrator should measure whether the ecosystem can preserve clinical context and generate shared knowledge.
- BMR should be presented as the first operational thread of IML, not as a narrow or isolated example.

PART VII

From Assessment to Action

A maturity framework becomes useful only when it helps an ecosystem decide what to improve next.

7.1 Why assessment is not enough

The previous parts defined the conceptual foundations of IML, described Health Information Ecosystems, clarified the meaning of interoperability and proposed a methodology for assessing maturity. However, assessment alone does not improve health.

A maturity score may describe the current state of an ecosystem, but it does not automatically create better care, better prevention or better public health. A framework that stops at measurement risks becoming another reporting exercise. IML is intended to avoid that limitation.

The purpose of IML is therefore not to produce a static label. Its purpose is to convert assessment into action. Each assessment should identify barriers, clarify priorities, propose realistic next steps and provide a basis for measuring progress over time.

This distinction is central to the philosophy of IML. Evaluation is not the destination. It is the beginning of a continuous improvement cycle.

7.2 The action logic of IML

IML proposes an action logic built around five successive questions. First, what information is needed to protect health in a specific use case? Second, where is this information currently generated? Third, which relationships prevent it from becoming trustworthy shared knowledge? Fourth, what changes are realistically achievable in the short term? Fifth, how will improvement be measured?

This sequence deliberately begins with a concrete health problem rather than with a technology. The AMR demonstrator illustrates this approach. The question is not whether a country uses a particular standard. The question is whether clinical, microbiological and public health information can be connected in a way that improves the understanding and management of resistant infections.

By applying the same reasoning to other clinical domains, IML can progressively move from assessment to implementation. The framework becomes a practical instrument for transformation rather than a descriptive dashboard.

7.3 Recommendations by maturity profile

Health information ecosystems differ widely. A country or organisation with limited digital infrastructure does not require the same recommendations as a highly digitalised but fragmented ecosystem. A single ranking would obscure these differences. For this reason, IML proposes maturity profiles rather than simple hierarchical classifications.

A maturity profile describes patterns of strengths and weaknesses. One ecosystem may have strong technical infrastructure but weak institutional engagement. Another may have limited resources but strong governance and professional trust. A third may possess advanced laboratory information systems but lack clinical context preservation.

Recommendations should therefore be adapted to the profile. The objective is not to tell every ecosystem to follow the same path. The objective is to identify the next realistic step that can produce measurable improvement.

Table 7.1 Suggested action logic by maturity profile

Maturity profile	Typical issue	Priority action
Low-resource or early-stage ecosystem	Limited infrastructure, fragmented records, weak digital continuity	Start with a narrow use case, minimal dataset and open standards.
Highly digitalised but fragmented ecosystem	Multiple systems, weak semantic alignment, difficult reuse of information	Map relationships, harmonise terminology and preserve clinical context.
Technically advanced but institutionally slow ecosystem	Standards exist but collaboration remains difficult	Strengthen institutional engagement, governance and shared accountability.
Strong public health infrastructure with limited clinical context	Surveillance receives data but not enough clinical meaning	Link laboratory, clinical and outcome information for selected use cases.
Mixed public-private payer ecosystem	Public and private payers operate independently despite caring for the same patients	Develop payer interoperability indicators and prevention-oriented data flows.

7.4 Supporting low-resource ecosystems

IML should be useful for countries and organisations with limited resources. Interoperability should not become a privilege reserved for highly financed health systems. On the contrary, open standards, minimal datasets and modular tools can help lower the threshold for progress.

In low-resource settings, the first objective should not be the immediate construction of a comprehensive national platform. A more realistic starting point is the identification of a high-value use case, such as AMR surveillance, maternal health, vaccination continuity or chronic disease follow-up. For each use case, a minimal dataset can be defined, the most important actors identified and a small number of information flows improved.

This approach allows progress without waiting for complete digital transformation. It also avoids the risk of importing large technological solutions that may not match local capacity, governance or clinical practice.

IML therefore proposes progressive interoperability: each step should be achievable, measurable and useful. A small improvement that genuinely supports clinical or public health decisions is more valuable than an ambitious architecture that cannot be implemented.

7.5 Supporting high-resource fragmented ecosystems

High-resource ecosystems face a different challenge. They may have extensive digital infrastructure, numerous national programmes, advanced hospitals, sophisticated payers and strong research capacity. Yet information may remain fragmented because systems were developed independently, governance is distributed and institutional incentives are not aligned.

In such contexts, the problem is rarely the absence of digital tools. It is the difficulty of transforming existing tools into a coherent Health Information Ecosystem. The ecosystem may contain many strong components but weak relationships between them.

IML should therefore help high-resource ecosystems identify fragmentation, duplicated infrastructures, semantic inconsistencies, insufficient clinical context, weak payer interoperability and institutional bottlenecks. The question is not whether these systems are digital. They are. The question is whether their digital maturity translates into knowledge continuity and better health.

7.6 The French paradox as an analytical profile

Some high-income countries illustrate what may be called the high-resource fragmentation paradox. They possess skilled professionals, advanced institutions, public financing mechanisms, research capacity and national digital health strategies. Yet their health information ecosystems may remain difficult to navigate for patients, clinicians and researchers.

France may be discussed as one example of this analytical profile, not as a judgement on a country but as an illustration of a broader pattern. Strong institutions and substantial resources do not automatically produce interoperable ecosystems. In some contexts, institutional complexity, professional segmentation, public-private boundaries and historical software fragmentation may limit the practical circulation of information.

The value of IML in such settings is to shift the discussion away from blame. The objective is not to ask why a country has failed. The objective is to identify which relationships require improvement and which use cases can create practical momentum.

7.7 The Danish reference without idealisation

Some countries are frequently cited as references because specific infrastructures have been well described in the scientific and institutional literature. Denmark, for example, is often discussed in relation to national microbiology data, antimicrobial resistance surveillance and healthcare data infrastructure.

Such references are useful, but they should not be idealised. A documented infrastructure may provide important lessons without becoming a universal model. No ecosystem is perfect. Each has strengths, limitations, institutional context and historical conditions that cannot simply be copied elsewhere.

IML should therefore use national references carefully. The purpose is not to designate champions but to identify transferable principles. A country can be exemplary in one domain and still raise open questions in another, including the practical conditions that facilitate external collaboration, responsiveness and shared learning.

7.8 Institutional inertia and engagement

Many interoperability projects encounter difficulties that are not primarily technical. Standards may exist. Legal frameworks may be defined. Pilot projects may be announced. Yet progress may remain slow because institutions do not engage, responsibilities are unclear, incentives are weak or collaboration depends on individuals rather than stable governance.

IML proposes treating this phenomenon as a research and assessment issue rather than as a political accusation. Institutional inertia should be studied, not denounced. It may result from workload, risk aversion, unclear mandates, competing priorities, lack of funding, legal uncertainty or absence of trusted collaboration mechanisms.

For this reason, institutional engagement and institutional readiness should become explicit components of interoperability maturity. An ecosystem is not mature simply because it has standards. It becomes mature when institutions are willing and able to collaborate around shared health objectives.

Institutional and generational lag should also be considered as part of interoperability maturity. This does not refer primarily to the biological age of decision-makers, but to the possible mismatch between the culture in which decision-making structures were formed and the digital, clinical and organisational realities they are now expected to govern.

In many health systems, senior decision-making structures were shaped before the current acceleration of digital health, artificial intelligence, data governance, interoperability standards, patient-mediated access and open-source software development. This may create a temporal gap between the urgency of present health information challenges and the institutional reflexes used to address them.

Such a gap may appear in several ways: slow responses to external scientific initiatives, absence of clear contact points, limited capacity to evaluate innovative proposals, reluctance to engage with open-source or university-led software models, and dependence on established procurement channels rather than independent quality evaluation. These patterns should not be interpreted as personal failure. They should be treated as observable features of institutional maturity.

For IML, responsiveness is not a courtesy. It is an interoperability capability. An ecosystem that cannot receive, examine, answer, redirect or openly reject a well-documented scientific proposal may struggle to transform digital health capacity into learning, correction and practical improvement.

IML therefore proposes that institutional responsiveness should be assessed explicitly. This includes the existence of public contact points, response procedures, scientific review pathways, mechanisms for engaging external proposals, transparency about why proposals are accepted or declined, and capacity to connect promising ideas with real-world experimentation.

This perspective is especially important in digital health, where innovation cycles are faster than traditional administrative cycles. Without mechanisms for timely review, institutions may unintentionally preserve outdated structures while the market of digital health software expands rapidly, often without sufficient independent evaluation of clinical usefulness, patient safety, interoperability quality or long-term public value.

7.9 Community building as an implementation strategy

Interoperability cannot be built by a single institution. It requires a community of practice bringing together clinicians, patients, laboratories, public health experts, payers, researchers, engineers, software developers, legal experts and policymakers.

The AMR demonstrator can serve as the first practical community-building mechanism for IML. It provides a recognised public health problem, a clear clinical need, measurable information flows and a concrete reason for different actors to collaborate.

A community should not be built around a score. It should be built around useful work. If IML can help actors improve a specific service, such as the clinical interpretation of resistant urinary tract infections, the framework will gain credibility through utility rather than promotion.

7.10 From pilot to reusable service

A pilot becomes valuable when it can be transformed into a reusable service. For IML, the AMR demonstrator should therefore not be designed as a one-time example. It should become a template for future use cases.

This requires a repeatable process: define the health problem, identify the actors, describe the required information, determine the minimal dataset, map information flows, assess maturity, propose improvements, implement a small service and measure results.

The same process could later be applied to vaccination continuity, chronic kidney disease, diabetes, cancer care pathways, rare diseases, medication safety or public health alerts. The AMR demonstrator is the starting point, not the limit of IML.

7.10.1 Institutional experimentation, reference implementation and digital health software quality

IML should not remain a repository of ideas, concepts or indicators. A maturity framework becomes scientifically and operationally useful only when it can be tested against real health information flows, real institutional constraints and real clinical or public health use cases.

This requirement is particularly important because digital health is not an ordinary software market. Health institutions are increasingly surrounded by numerous applications, platforms, dashboards, artificial intelligence modules, data services and integration tools. Some may be certified, accredited, procured or commercially adopted within specific frameworks. However, market adoption, declared functionality, certification or procurement status do not necessarily demonstrate that a solution preserves clinical meaning, supports correction, remains auditable, avoids lock-in, contributes to continuity of care or improves health information usefulness at ecosystem level.

In digital health, software quality is not only a technical matter. It is a patient safety issue. Digital systems may influence clinical decisions, continuity of care, public health surveillance, patient rights, professional workflows and institutional accountability. Failures in information quality, interoperability, auditability, correction or portability may affect diagnosis, treatment, prevention and ultimately human lives.

For this reason, IML requires more than conceptual assessment. It requires independent, transparent and reproducible methods for evaluating the quality of digital health software from a health-oriented perspective. The question should not be limited to whether a product exists, whether it is technically functional or whether it has been adopted by an institution. The deeper question is whether it can be trusted in a field of extreme sensitivity.

IML should therefore evolve toward a reference software implementation: a universal, open-source, vendor-neutral and academically governed environment that can serve as a public benchmark for testing health information interoperability in practice. Universal does not mean uniform or imposed on every health system. It means that the reference implementation should be based on open principles, reusable components and transparent methods that can be adapted to different countries, institutions and use cases.

Its purpose would not be to replace existing clinical software, hospital systems, laboratory systems, payer platforms or national infrastructures. Its purpose would be to provide a shared reference point in the middle of fragmentation: a controlled environment where IML indicators, datasets, identity and consent mechanisms, auditability, correction procedures, clinical context preservation, public health reuse, payer interoperability and responsible AI readiness can be tested together.

Such a reference implementation would also allow comparison with market solutions. Public, private, open-source or commercial tools could be assessed not only by their declared functionality, certification status or adoption level, but by their ability to perform against transparent IML criteria: preservation of meaning, clinical context, interoperability across systems, traceability, reversibility, correction, security, portability, public health value, payer interoperability and measurable usefulness.

For this reason, the next stage of IML should combine three elements: an academic or public health reference environment, a concrete field of experimentation, and an open-source reference software implementation. These elements should not be separated. Scientific review without field testing may remain abstract. Field testing without methodological discipline may remain local and difficult to reproduce. Software development without governance may create yet another isolated tool.

The appropriate model is close to a learning health system or academic health campus logic: research, care, public health, software development and evaluation should be organised in the same institutional environment or through a formally governed partnership. In such a setting, IML concepts could be reviewed scientifically, translated into indicators, implemented in software, tested against real information pathways, compared with existing digital health tools and revised according to observed results.

The long-term objective should therefore include an open-source reference implementation capable of testing IML methods, documenting datasets, preserving clinical context, supporting auditability, enabling correction, managing identity and consent, producing reproducible indicators and demonstrating how assessment can become action. This reference implementation should not be presented as a production platform at the outset. It should begin as a research, comparison and validation environment, progressively tested under institutional, ethical, legal and cybersecurity governance.

At this stage, IML does not host an operational database and does not provide a production software platform. Database hosting and software deployment should be treated as future institutional objectives, conditional on formal partnership, ethical and legal review, data governance, cybersecurity safeguards, transparent access rules, correction procedures, open-source maintenance and sustainability planning.

IML can begin as a manuscript, but it can mature only if ideas become tested methods, methods become open-source reference software, and software is evaluated against real health information flows and compared with the fragmented market of digital health solutions.

7.11 Roadmap for early implementation

The first implementation phase should remain deliberately modest. IML should begin with a narrow, clinically meaningful service. For example, in the AMR use case, the first service could help distinguish resistant isolates from clinically significant infections by linking microbiology, symptoms, diagnosis, treatment and outcome information.

A second phase could extend the service to public health surveillance, allowing aggregated and privacy-preserving analysis of clinically meaningful resistant infections. A third phase could explore payer information, care pathways and prevention opportunities. A fourth phase could evaluate AI support once the underlying information flows are sufficiently reliable.

This roadmap reflects a principle that should guide IML: implementation should follow maturity. Artificial intelligence, advanced analytics and large-scale dashboards should not precede trustworthy information flows. They should be built upon them.

7.12 Measuring improvement over time

Continuous improvement requires repeated measurement. An initial maturity assessment should create a baseline. Subsequent assessments should examine whether specific barriers have been reduced, whether information flows have improved and whether decisions have become better supported.

Progress may be measured through process indicators, such as the proportion of laboratory results linked to clinical context, and outcome-oriented indicators, such as reduced duplicated testing, improved antimicrobial stewardship or better surveillance of clinically meaningful infections.

IML should be cautious when attributing health outcomes directly to interoperability. Many factors influence outcomes. Nevertheless, the framework can measure whether the conditions required for better decisions are improving.

7.13 Transition to the Open Clinical Workspace

The movement from assessment to action naturally raises a practical question. Which tools can help ecosystems improve without replacing all existing systems? This question leads to the concept of an Open Clinical Workspace.

An Open Clinical Workspace should not be conceived as another closed electronic health record. It should be a modular environment capable of interacting with existing systems, importing information on request, preserving clinical context and supporting collaborative knowledge production under human responsibility.

The following part therefore moves from action planning to implementation tools. It examines how an open, standards-based workspace could help translate the IML framework into practical services for clinicians, public health authorities, payers, researchers and patients.

Key Messages - Part VII

- Assessment is useful only when it leads to action.
- IML should guide continuous improvement rather than produce static rankings.
- Different ecosystems require different recommendations according to their maturity profiles.
- Low-resource ecosystems and high-resource fragmented ecosystems both need support, but not the same kind of support.
- Institutional inertia should be studied as an interoperability issue rather than treated as a political accusation.

- The AMR demonstrator can become the first operational thread for community building and practical implementation.
- The next logical step is the Open Clinical Workspace, conceived as an implementation tool rather than a replacement system.
- Institutional responsiveness should be treated as an interoperability capability: ecosystems need clear pathways to receive, review and test external scientific and software proposals.

PART VIII

Open Clinical Workspace

The previous part established that assessment alone is not sufficient. A maturity framework becomes useful only when it helps ecosystems move from diagnosis to practical action. This part introduces the Open Clinical Workspace as a proposed implementation instrument for IML: not a replacement for existing systems, but a practical environment designed to connect, contextualise and use trustworthy health information under human responsibility.

8.1 Why an Open Clinical Workspace?

Many interoperability initiatives remain limited because they stop at the level of assessment, policy or technical architecture. They identify fragmentation but do not always provide a practical environment in which clinicians, laboratories, payers, public health actors and researchers can experience the benefits of better information continuity.

IML therefore proposes that a maturity framework should progressively be linked to an implementation tool. The Open Clinical Workspace is conceived as this bridge. Its purpose is to translate the principles of Health Information Ecosystems into a usable, modular and open-source clinical environment.

The Open Clinical Workspace should begin with a limited use case rather than an all-purpose platform. The antimicrobial resistance demonstrator provides a suitable starting point because it requires microbiology, clinical context, treatment information, public health interpretation and governance. In such a setting, the workspace can demonstrate the practical value of IML without pretending to solve every interoperability challenge at once.

The Open Clinical Workspace should be understood as the future software expression of the IML framework. It should not be another proprietary platform added to an already fragmented market. It should be an open-source, vendor-neutral and academically governed reference environment in which IML methods can be tested, criticised and compared with existing digital health solutions through real use cases.

Its purpose is not only to support implementation, but also to provide a practical benchmark for digital health software quality. In a market where many tools claim to support interoperability, artificial intelligence, analytics, coordination or patient engagement, IML should help ask a more demanding question: does the software preserve meaning, context, trust, auditability, correction capacity, portability and measurable usefulness across Health Information Ecosystems?

8.2 An Open-Source Reference Platform for Connected and Underserved Ecosystems

The Open Clinical Workspace should not be understood as another closed, proprietary or vendor-dependent electronic health record. Its long-term objective is to become an open-source, vendor-neutral reference environment capable of operating in two complementary situations: connecting and contextualising information from existing systems where digital infrastructure is already present, and providing a progressively deployable clinical and public health foundation where usable digital systems are absent, fragmented, unaffordable or unable to exchange meaningful information.

This dual purpose responds to a major global inequality. In 2025, approximately 6 billion people, or 74 per cent of the world population, were using the Internet, while 2.2 billion remained offline. Mobile connectivity figures reveal an even broader divide: 3.4 billion people were not using mobile Internet in 2024. Around 300 million lived outside mobile broadband coverage, while 3.1 billion lived within coverage but did not use it because of barriers such as device affordability, skills, literacy, trust or relevant services. Ninety-three per cent of people who remained unconnected lived in low- and middle-income countries. Within those countries, rural adults were 25 per cent less likely than urban adults to use mobile Internet, and women were 14 per cent less likely than men to do so.

Digital exclusion is not limited to connectivity. The global electricity access rate remained approximately 92 per cent in 2024, with rural communities in sub-Saharan Africa experiencing a particularly large and persistent deficit. At the same time, more than 800 million people lacked official proof of identity, and 2.8 billion people did not have access to a digital identity suitable for online transactions. These populations overlap and must not be added together as a single total. They nevertheless demonstrate that digital health cannot safely be designed around permanent connectivity, personal smartphones, stable electricity and universal digital identification.

The Open Clinical Workspace should therefore be designed according to principles of inclusive and progressive digitalisation. It should support local hosting, intermittent connectivity, offline-first workflows, delayed synchronisation, multilingual interfaces, modest hardware, low-bandwidth exchange, accessible user interfaces and migration from paper or partially digital records. It should not require every patient to possess a smartphone, permanent Internet connection or online digital identity. Existing national, local or organisational identifiers should be used through governed identity and access mechanisms without creating a new universal identity number.

In digitally mature environments, the workspace should function as an interoperability, clinical-context and quality layer. It should connect with hospital systems, primary care software, laboratory platforms, pharmacy applications, payer systems and public health infrastructures through APIs, FHIR resources, terminology services, structured files or other appropriate exchange mechanisms. It should import relevant information on demand, preserve the context that gives information meaning, support correction and auditability, and return structured outputs to authorised systems.

In underserved environments, the same reference implementation should be capable of providing a minimal but usable digital foundation. Its initial modules could include patient registration, essential clinical documentation, laboratory requests and results, treatment recording, outcome follow-up, correction mechanisms and aggregated public health reporting. The first implementation should remain focused on the AMR/BMR demonstrator, connecting microbiology, symptoms, diagnosis, treatment and outcomes for multidrug-resistant urinary tract infections. Additional clinical and public health modules could then be added progressively according to local priorities and resources.

IML should not rebuild functions that are already available in mature open-source digital health projects. The Open Clinical Workspace should instead evaluate, reuse, connect and, where necessary, extend existing open-source components. IML-specific development should concentrate on the areas that distinguish the framework: Clinical Context Preservation, Continuity of Health Information, correction across systems, evidence grading, institutional responsiveness, payer interoperability, professional role alignment, digital health software quality benchmarking and learning from measured outcomes.

The Open Clinical Workspace should also serve as an independent reference environment for evaluating existing digital health software. Commercial, public and open-source solutions could be tested against transparent health-oriented criteria, including preservation of meaning, clinical usefulness, auditability, correction capacity, security, resilience, portability, reversibility, accessibility, offline operation and long-term maintainability.

Open source should not be confused with cost-free implementation. Even when software has no licensing fee, deployment requires equipment, electricity, configuration, translation, training, cybersecurity, technical support, governance and sustainable maintenance. The purpose of an open-source reference implementation is not to eliminate these costs, but to reduce avoidable licensing dependency, prevent vendor lock-in, enable local adaptation, preserve institutional autonomy and allow countries to build upon shared public infrastructure rather than repeatedly purchasing isolated systems.

The guiding principle should therefore be:

> **Connect where systems exist. Provide a practical and inclusive starting point where they do not.**

8.3 Interacting with Existing Software

A central requirement of the Open Clinical Workspace is the ability to interact with existing applications without forcing institutions to abandon them. Hospitals, laboratories, payers and clinicians cannot realistically replace their information systems simply because a new interoperability framework has been proposed.

The workspace should therefore be designed around connectors, interfaces and import mechanisms. In its simplest form, it may begin by importing structured files, laboratory extracts or manually validated datasets. In more mature environments, it may interact through APIs, FHIR resources, terminology services or event-based messaging.

The guiding principle is pragmatic interoperability: use the most mature exchange method available in a given ecosystem, while encouraging gradual progression toward open, standardised and semantically consistent exchange.

8.4 Import on Demand

IML proposes import on demand as an important design principle. Health information should not circulate permanently or indiscriminately simply because technology makes it possible. Instead, information should be accessed, imported or synchronised according to a defined clinical, public health or research purpose.

Import on demand supports proportionality. It allows the workspace to retrieve relevant information for a specific question while limiting unnecessary duplication, reducing privacy risks and respecting local governance constraints.

For the AMR demonstrator, import on demand could allow a clinician or surveillance team to bring together microbiology results, clinical symptoms, urinary findings, treatment information and outcome data only when a suspected multidrug-resistant infection needs to be assessed.

8.5 Open Standards and Vendor Neutrality

The Open Clinical Workspace should be vendor-neutral. Its purpose is not to create dependency on a specific commercial platform or technology provider. It should be compatible with open standards, open terminologies and transparent governance mechanisms whenever possible.

Relevant standards may include HL7 FHIR for exchange, openEHR for clinical modelling, SNOMED CT and LOINC for terminology, ICD for statistical classification, DICOM for imaging and established security standards for authentication and access control. The exact combination should depend on local maturity and use case requirements.

Vendor neutrality does not mean opposition to industry. On the contrary, industry may contribute valuable tools, devices and services. The principle is that industrial innovation should strengthen ecosystem interoperability rather than create new silos.

8.6 Modular Architecture

The workspace should be modular. A modular architecture allows different components to evolve independently and makes implementation possible in low-resource as well as high-resource settings.

A minimal architecture could include a data import module, terminology service, clinical context module, user interface, consent and access management layer, audit log, analytics layer and export module. More advanced versions could include AI-assisted summarisation, decision support and public health dashboards.

A relational database such as PostgreSQL may provide a practical foundation for early prototypes because it is mature, open-source and widely supported. However, IML should avoid locking the architecture to a single database technology. The methodological principle matters more than the initial implementation choice.

8.7 Clinical Context Layer

The clinical context layer is one of the most important components of the proposed workspace. Interoperability that transfers data without context may increase confusion rather than improve care.

The workspace should therefore preserve the relationship between observations and the clinical circumstances that give them meaning. A microbiological result should be linked, when available, to symptoms, sampling context, suspected diagnosis, contamination risk, treatment decision and outcome. A prescription should be linked to its indication. A discharge summary should be linked to follow-up requirements.

This layer is particularly important for antimicrobial resistance. A resistant isolate does not automatically represent a symptomatic infection. The workspace should help distinguish colonisation, contamination, asymptomatic bacteriuria and clinically meaningful infection.

8.8 Identity, Consent and Trust

A workspace that connects information across organisations must be built around identity, consent and trust. Users must know whose information is being accessed, under what authority, for which purpose and with what traceability.

Digital identity should not be reduced to technical authentication. It is also a governance issue. Patients, clinicians, organisations and software agents must be represented in ways that support accountability, auditability and legal certainty.

Consent mechanisms should reflect the sensitivity of the information and the context of use. Some uses may rely on direct care relationships, others on public health authority, research governance or anonymisation. The workspace should make these distinctions visible rather than hiding them behind a single technical access model.

The workspace should also implement the identity-access layer described in Domain 3. It should avoid unnecessary storage or exposure of national identifiers and should support purpose-limited access tokens, patient-mediated authorisation, trusted identity assertions, auditability, revocation and correction. QR codes or mobile applications may be used as access mechanisms only when they carry temporary, signed and revocable tokens rather than sensitive identity or health data in clear text.

8.9 Hardware, Operating Systems and Interfaces

A practical interoperability framework must also consider the material conditions under which information systems operate. A Health Information Ecosystem is not composed only of applications, databases and exchange standards. It also depends on workstations, servers, embedded devices, mobile equipment, networks, peripheral devices, user interfaces and operating systems.

Operating-system diversity is a real feature of healthcare. Clinical environments may still depend on Microsoft Windows and, in some legacy settings, DOS-derived workflows; they may also include macOS, whose technical lineage is related to UNIX and BSD traditions, as well as UNIX systems, GNU/Linux distributions and BSD systems such as FreeBSD. This diversity should not be treated as a curiosity or a secondary technical detail. It affects maintainability, cybersecurity, portability, device compatibility, long-term support, user experience and the capacity of institutions to keep essential services working over time.

IML should not compare operating systems by asking which brand is intrinsically superior. That question would be too abstract for health. The relevant comparison is use-case based: does the operating environment support the clinical, public health or research function safely, sustainably and interoperably in the context where it is deployed? The reference point should therefore be a neutral quality model rather than a preferred platform. For future referenced versions, ISO/IEC 25010:2023 can provide the general product quality vocabulary, POSIX.1-2024 can inform portability expectations for UNIX-like environments, and security baselines such as the NIST Cybersecurity Framework 2.0 and CIS Benchmarks can inform hardening, governance and risk management.

The comparison protocol should begin with an inventory of the operating environments actually used in the ecosystem, including version, support status, patch policy, hardware dependence and connection to clinical devices. It should then test representative interoperability tasks, such as secure authentication, import on demand, laboratory-result exchange, audit logging, terminology-service access, encrypted storage, backup and recovery, and export to authorised care, public health or research users. The protocol should also examine lifecycle sustainability, because an operating system that works today but can no longer be patched, documented or maintained may become a future source of clinical and institutional risk.

The quality criteria should remain simple and comparable across platforms. IML should assess reliability, security, maintainability, portability, interoperability support, auditability, performance under real clinical constraints, compatibility with essential hardware and peripheral devices, accessibility for users, ease of administration, documentation, recovery capacity and institutional autonomy. These criteria do not require IML to catalogue every clinical software product. They require IML to ask whether the operating environment on which healthcare depends is sufficiently safe, durable and interoperable to support trustworthy information.

This perspective is particularly important for the Open Clinical Workspace. The workspace should remain lightweight, portable and progressively deployable. It should function on modest infrastructure when necessary, while remaining compatible with more advanced environments. Its value should not depend on a single operating system family. Its design should encourage portability, documented dependencies and reproducible deployment across heterogeneous health information ecosystems.

8.10 Database Quality and Health Information Ecosystems

Health Information Ecosystems depend not only on operating systems and interfaces, but also on the databases that preserve, organise, query, correct and transmit health information. The diversity of database technologies is considerable: relational databases, document stores, graph databases, time-series databases, embedded databases, distributed databases and proprietary clinical data stores may all coexist inside the same ecosystem. IML should not compare these technologies by brand preference or market dominance. The relevant question is whether a database environment can support trustworthy health information over time.

Healthcare provides a particularly demanding paradigm for database-quality comparison. It requires integrity because clinical information must not be silently lost or corrupted; traceability because access, correction and reuse must be auditable; security because health information is sensitive; resilience because care must continue during incidents; portability and reversibility because institutions should not become prisoners of opaque formats; and semantic consistency because data must preserve clinical meaning across settings.

A database-quality assessment should therefore be use-case based. For the AMR/BMR demonstrator, the same minimal dataset for MDR E. coli urinary tract infection could be implemented across different database approaches. The comparison should examine import, validation, queryability, linkage between microbiology and clinical context, correction, audit logs, export, anonymisation or pseudonymisation, backup, restoration, migration and long-term maintainability. The purpose is not to decide which database is universally best. The purpose is to determine which database environment is fit for a specific health information task under real governance constraints.

Neutral references can support this assessment. ISO/IEC 25010:2023 provides a general model for software and ICT product quality. ISO/IEC 9075:2023 provides a reference for SQL-based database language portability. NIST SP 800-53 Rev. 5 provides a catalogue of security and privacy controls for information systems and organisations. CIS Benchmarks provide practical configuration guidance for securing many database products. IML can use these references without reducing the assessment to compliance alone; the health use case remains the final test.

The proposed comparison protocol is deliberately simple. First, define the health use case and the minimum clinical dataset. Second, document the database technologies used for priority health information. Third, test whether the data model preserves clinical context, auditability, correction and export. Fourth, evaluate backup, restoration, migration and secure configuration. Fifth, compare the capacity of each environment to support care, public health, research and learning without creating avoidable dependency. In this way, health becomes a rigorous benchmark for database quality.

8.11 Artificial Intelligence Under Human Responsibility

Artificial intelligence may become useful within the Open Clinical Workspace, but only if its role is clearly defined. AI should assist professionals by summarising information, detecting inconsistencies, suggesting missing context, identifying risk patterns or supporting research extraction. It should not replace clinical judgement or institutional accountability.

The quality of AI output depends on the quality of the underlying information. Fragmented, incomplete or poorly contextualised data will produce unreliable AI assistance. The workspace should therefore treat AI readiness as a consequence of information maturity rather than as a shortcut around it.

Every AI-assisted function should remain transparent, auditable and subordinate to human responsibility. In the context of AMR, AI may help identify probable clinically meaningful infections, but final interpretation must remain clinical and microbiological.

8.12 Minimum Viable Prototype

The first prototype should be deliberately modest. A minimum viable Open Clinical Workspace for the AMR demonstrator could focus on multidrug-resistant Escherichia coli urinary tract infections.

Its initial functions could include importing microbiology results, linking them to minimal clinical context, classifying cases according to infection likelihood, documenting treatment, recording outcome and producing aggregated indicators for quality improvement and surveillance.

The prototype should not attempt to solve national interoperability. It should demonstrate that a specific clinical problem can be improved by connecting the right information in the right context. Success would be measured not by the number of connected systems but by the quality of the decisions supported.

8.13 From Prototype to Reusable Service

If the initial prototype proves useful, the same design logic could be applied to other clinical and public health use cases: vaccination continuity, chronic disease follow-up, hospital discharge coordination, cancer pathway monitoring, medication safety, rare disease registries or emergency care summaries.

The value of the Open Clinical Workspace lies in its reusability. Each new use case should not require a completely new architecture. Instead, the workspace should provide reusable modules for identity, consent, terminology, import, context preservation, audit, analytics and export.

This approach transforms implementation into cumulative learning. Every use case strengthens the common infrastructure for future use cases.

8.14 Governance of the Workspace

The Open Clinical Workspace cannot be governed as a purely technical project. Its governance should include clinicians, patients, microbiologists, public health representatives, payers, informaticians, security experts, legal experts and open-source contributors.

Governance should define priorities, validate use cases, approve data models, supervise security, evaluate risks, document decisions and ensure that the workspace remains aligned with the purpose of protecting health.

Open-source governance is particularly important. Transparency of code does not automatically guarantee trust. Trust requires documentation, review, accountability, maintenance and a community capable of correcting errors over time.

Because digital health software may influence patient safety, public health decisions and institutional accountability, governance of the Open Clinical Workspace should include explicit software-quality evaluation. Open-source availability alone is not sufficient. The workspace should be governed through documented testing protocols, transparent issue tracking, reproducible deployment, cybersecurity review, clinical safety assessment, correction procedures, version control and independent scientific review.

8.15 Transition to Responsible AI and Scientific Integrity

The Open Clinical Workspace naturally leads to a broader question. If health information becomes more connected, more contextualised and more reusable, artificial intelligence will increasingly be able to assist interpretation, analysis and decision support.

This creates both opportunity and responsibility. The next part therefore examines responsible artificial intelligence and scientific integrity. It clarifies how AI can contribute to IML while preserving transparency, human responsibility and independence from any specific technology provider.

8.16 Proposed Modules of the Open Clinical Workspace

- Data import: Retrieve or receive relevant information from existing systems. Initial AMR use case: import microbiology results and minimal clinical context.
- Terminology service: Support semantic consistency. Initial AMR use case: map organisms, antibiotics, specimens and diagnoses.
- Clinical context layer: Preserve the meaning of observations. Initial AMR use case: link culture results to symptoms, diagnosis and treatment.
- Identity and consent: Manage authorised access and accountability. Initial AMR use case: identify patient, clinician, organisation and purpose of use.
- Audit log: Record access, changes and decisions. Initial AMR use case: trace who accessed or modified AMR case information.
- Analytics layer: Generate indicators and dashboards. Initial AMR use case: estimate clinically meaningful MDR E. coli UTI burden.
- Database and storage quality: Preserve, query, audit, correct, export, back up and restore health information without avoidable dependency. Initial AMR use case: store and retrieve microbiology, clinical context, treatment and outcome information with traceable correction and export.
- AI assistance: Support summarisation and pattern recognition under human responsibility. Initial AMR use case: flag missing context or suggest case review.
- Export module: Return structured outputs to authorised users or systems. Initial AMR use case: produce aggregate indicators for quality improvement and surveillance.

Key Messages - Part VIII

- The Open Clinical Workspace is an implementation bridge between maturity assessment and practical action.
- It should not replace existing systems; it should interact with them.
- Import on demand supports proportionality, privacy and practical use.
- The clinical context layer is essential for preserving the meaning of health information.
- The workspace should be open-source, modular, vendor-neutral and progressively deployable.
- Operating systems should be assessed through neutral quality criteria, including security, maintainability, portability, reliability, auditability and support for interoperable clinical use.
- Database technologies should be assessed through health-oriented quality criteria, including integrity, auditability, security, reversibility, portability, resilience and clinical context preservation.
- Artificial intelligence should assist interpretation under human responsibility, not replace judgement.
- The AMR demonstrator provides a realistic starting point for a minimum viable prototype.
- Digital health software quality is a patient safety issue; IML should provide an open-source, academically governed reference environment for testing whether digital health tools preserve meaning, context, trust, auditability, correction capacity, portability and measurable usefulness.
- Identity mechanisms in the Open Clinical Workspace should rely on purpose-limited access tokens, patient-mediated authorisation and auditable trust mechanisms rather than a new universal identity number.

PART IX

Responsible Artificial Intelligence and Scientific Integrity

Artificial intelligence now participates in the production, organisation, interpretation and communication of health information. For this reason, it cannot be treated as an external topic added at the end of a digital health strategy. It belongs inside the Health Information Ecosystem itself.

The question for IML is therefore not whether artificial intelligence should be used in health. It is under which conditions AI can contribute to better health while preserving scientific integrity, clinical responsibility, patient trust and democratic governance.

9.1 Why responsible AI belongs in IML

IML begins from a simple principle: health information should protect health. Artificial intelligence can support this objective when it helps professionals and institutions transform fragmented information into usable knowledge. It can also create risk when it amplifies poor-quality data, obscures uncertainty, reproduces bias or creates an illusion of authority.

Responsible AI is therefore not a separate ethical appendix. It is a maturity requirement. A Health Information Ecosystem that cannot govern AI responsibly cannot be considered mature, even if its technical infrastructure appears advanced.

9.2 AI as an ecosystem actor

Artificial intelligence should be understood as an ecosystem actor, but not as an autonomous moral or clinical agent. AI systems may summarise, classify, detect patterns, translate, retrieve evidence, identify missing context or support decision-making. Their contribution depends on the quality of the information they receive and the governance under which they operate.

In a mature ecosystem, AI does not replace clinicians, researchers or public authorities. It extends their capacity to analyse information while remaining accountable to human judgement, institutional responsibility and legal frameworks.

9.3 From information processing to knowledge support

The value of AI in health does not lie in producing more outputs. Its value lies in supporting better knowledge. An AI system that generates a plausible summary from incomplete information may be less useful than a simpler tool that identifies missing data, preserves clinical context or alerts the user to uncertainty.

IML therefore proposes evaluating AI not by sophistication alone but by its contribution to trustworthy knowledge. The central question becomes: does the AI function help people make better decisions, or does it merely automate information processing?

9.4 Human responsibility

Clinical judgement remains human. Scientific interpretation remains human. Ethical responsibility remains human. AI may assist, but it cannot assume professional accountability.

This principle is particularly important in clinical contexts where information is incomplete, ambiguous or value-laden. A model may identify correlations; a clinician must interpret their meaning for a specific patient. A system may generate a recommendation; an institution must determine whether and how such recommendations may be used.

IML therefore considers human responsibility a non-negotiable condition for AI use in health information ecosystems.

9.5 Transparency of AI-assisted writing and analysis

This manuscript itself was prepared with editorial and analytical assistance from OpenAI ChatGPT. This use should be disclosed because transparency is part of scientific integrity. AI supported drafting, structuring, language refinement and conceptual exploration. It did not replace authorship, accountability or final judgement.

The author remains responsible for the conceptual choices, methodological orientation, scientific claims and final content of the manuscript. Future IML publications should continue this practice by disclosing meaningful AI assistance when it contributes substantially to writing, analysis or interpretation.

9.6 Inspiration from the OpenAI Charter

The development of IML has been inspired in part by the aspiration expressed in the OpenAI Charter that advanced artificial intelligence should benefit all of humanity. This reference is not a claim of endorsement, partnership or institutional affiliation. It is an ethical inspiration.

Within IML, this aspiration is translated into a practical health objective: AI should contribute to trustworthy information, better decisions, improved prevention, scientific collaboration and more equitable Health Information Ecosystems. It should not increase opacity, dependency, inequality or institutional fragmentation.

9.7 Privacy, proportionality and data protection

Health data are among the most sensitive categories of information. Responsible AI therefore requires privacy by design, data minimisation, proportionality, access control, auditability and compliance with applicable legal frameworks.

The objective is not to feed AI systems with all available data. The objective is to provide appropriate information for clearly defined purposes, under explicit governance. In this respect, the principle of import on demand proposed for the Open Clinical Workspace is also relevant to AI: the least necessary information should be used for the most clearly justified purpose.

9.8 Bias, equity and representativeness

AI systems may reproduce or amplify inequalities when trained or deployed on incomplete, biased or poorly representative data. In health, this risk is particularly serious because vulnerable populations are often those whose information is most fragmented, least structured or least visible within digital systems.

IML therefore links responsible AI to health equity. AI readiness should include the capacity to evaluate bias, document limitations, monitor performance across populations and ensure that AI does not reinforce existing disparities in access, diagnosis, treatment or follow-up.

9.9 Scientific integrity and reproducibility

AI-assisted analysis must remain compatible with scientific integrity. This requires traceability of methods, documentation of data sources, version control of tools, explicit distinction between evidence and hypothesis, and reproducibility whenever possible.

Where AI supports research, evaluation or policy recommendations, its role should be described. Where AI generates outputs used in clinical or public health decisions, those outputs should be auditable, contestable and interpreted by qualified professionals.

9.10 AI readiness as a maturity dimension

IML proposes that AI readiness should be considered a cross-cutting dimension of maturity. AI readiness does not simply mean access to advanced models. It means the ability of an ecosystem to deploy AI safely, transparently and usefully.

Potential indicators include the existence of AI governance policies, clinical validation procedures, audit mechanisms, bias monitoring, human oversight, transparency requirements, incident reporting, cybersecurity controls and training for professionals.

9.11 Boundaries of AI in health information ecosystems

Not every task should be automated. Not every dataset should be reused. Not every prediction should become a recommendation. A mature ecosystem must be capable not only of using AI but also of limiting its use.

The ability to say no is part of maturity. Responsible AI requires judgement about appropriate scope, acceptable risk, clinical relevance and social legitimacy.

9.12 Transition to roadmap and future work

Responsible AI reinforces the central logic of IML. Health is the objective. Information is the foundation. Interoperability is the path. AI may assist this path, but only if it remains governed by human responsibility, scientific evidence and public trust.

The final part of this manuscript therefore turns from principles and frameworks toward a roadmap for future work: scientific review, methodology refinement, technical implementation, case studies, publication strategy and community building.

Key Messages - Part IX

- Artificial intelligence is an ecosystem actor, not an autonomous clinical or moral authority.
- AI should be evaluated by its contribution to trustworthy knowledge and better decisions, not by sophistication alone.
- Human responsibility remains essential in clinical judgement, scientific interpretation and ethical decision-making.
- Transparency of AI-assisted writing and analysis is part of scientific integrity.
- The OpenAI Charter aspiration to benefit all of humanity is used as ethical inspiration, not as endorsement or affiliation.
- Privacy, proportionality and data minimisation are necessary conditions for responsible AI in health.
- AI readiness should be assessed as a cross-cutting maturity dimension within IML.
- The capacity to limit inappropriate AI use is part of ecosystem maturity.

PART X

Roadmap and Future Work

"A framework becomes useful only when it helps people move from shared understanding to practical improvement."

10.1 Why a roadmap matters

A scientific framework cannot remain only a set of concepts. If it is to serve health, it must become a pathway for action. The preceding parts of this manuscript have defined the philosophical, conceptual, methodological and operational foundations of the Interoperability Maturity Lab. The final part describes how these foundations may evolve into a practical programme of work.

The purpose of this roadmap is not to impose a single model on every country or institution. Health systems differ in their histories, financing structures, legal frameworks, technological infrastructures and professional cultures. A roadmap for interoperability must therefore be progressive, adaptable and respectful of context.

IML proposes that future development should follow a simple sequence: clarify concepts, validate methodology, demonstrate practical usefulness, build reusable tools, invite external review and maintain transparent versioning. This sequence is intentionally modest. It does not promise transformation overnight. It proposes a disciplined way to advance.

10.2 From manuscript to working programme

The present manuscript should be understood as the founding document of IML, not as its final form. Its immediate objective is to define a shared language and a coherent intellectual architecture. Future work should transform this architecture into specific instruments: indicator sets, assessment templates, case study protocols, software prototypes, governance recommendations and implementation guides.

This progression is important because many digital health initiatives fail when the conceptual, methodological and operational layers are developed separately. A technology may be built before the problem is clearly defined. An assessment may be performed before indicators are validated. A pilot may be launched before governance is understood. IML seeks to avoid this fragmentation by maintaining continuity between theory, methodology and implementation.

10.3 Versioning as scientific discipline

IML should evolve through explicit versioning. Each version of the manuscript, methodology and associated tools should be traceable. Changes should be documented. Concepts should not disappear silently. Indicators should not be modified without explanation. This approach treats the framework as a living scientific artefact rather than a static report.

Versioning serves several purposes. It allows external reviewers to comment on a specific state of the work. It enables comparison between methodological iterations. It supports transparency when indicators, scoring systems or definitions evolve. It also protects the intellectual continuity of the project.

The Alpha phase should focus on conceptual consolidation. The Beta phase should invite external scientific review. A release candidate should integrate comments, clarify limitations and prepare the framework for public use. Later versions should incorporate evidence generated by pilots and case studies.

10.4 External scientific review

The credibility of IML will depend on its capacity to invite criticism. External review should not be considered a ceremonial step at the end of the process. It should become a structural component of the framework.

Reviewers may include clinicians, health informatics researchers, public health experts, interoperability specialists, representatives of payer organisations, patient representatives, software developers and policymakers. Their perspectives will differ, and this diversity is valuable. A Health Information Ecosystem cannot be understood from a single professional viewpoint.

External review should address at least four questions: whether the concepts are clear, whether the proposed dimensions are relevant, whether the methodology is feasible and whether the framework can support practical improvement. Review should also identify omissions, risks and possible unintended consequences.

10.5 Methodology handbook

A separate methodology handbook should be developed after the conceptual framework has stabilised. This handbook would translate the principles of IML into operational procedures. It should define indicators, evidence categories, scoring rules, weighting options, reviewer roles, documentation requirements and validation procedures.

The handbook should distinguish between core indicators, optional indicators and context-specific modules. Core indicators would allow comparability over time. Optional indicators would support adaptation to specific settings. Context-specific modules could address antimicrobial resistance, chronic disease management, payer interoperability, artificial intelligence readiness or low-resource implementation.

This separation between the founding manuscript and the methodology handbook is important. The manuscript defines the why and the what. The handbook defines the how.

10.6 Technical handbook

A technical handbook should accompany the methodology handbook. Its purpose would not be to prescribe a single technology stack, but to describe reference architectures, interoperability patterns and implementation principles compatible with open standards.

Topics may include FHIR-based exchange, terminology services, openEHR archetypes, patient identity, consent management, audit trails, secure APIs, import-on-demand mechanisms, modular databases, clinical context layers and integration with existing software. The technical handbook should also address hardware, operating-system diversity and interface constraints, using neutral quality criteria rather than platform preference, because interoperability is affected by the entire digital environment in which healthcare is practised.

The technical handbook should remain vendor-neutral. Its purpose is to help ecosystems avoid dependency, duplication and fragmentation, not to promote a particular product.

10.7 Case study programme

Case studies are essential because interoperability becomes meaningful only when applied to concrete problems. IML proposes antimicrobial resistance and multidrug-resistant bacteria as the first demonstrator, but future case studies should progressively broaden the field.

Potential case studies include chronic disease management, vaccination continuity, laboratory result interpretation, cross-border care, emergency care, medication reconciliation, payer interoperability, rare disease registries, environmental health and responsible AI deployment.

Each case study should be designed to answer a precise question. It should identify the information required, the actors involved, the current fragmentation points, the maturity dimensions tested and the improvement expected. In this sense, case studies become laboratories for the framework itself.

10.8 Implementation pathway

Implementation should proceed in stages. The first stage consists of conceptual consolidation and expert review. The second stage consists of methodological formalisation. The third stage consists of a small demonstrator focused on a specific use case. The fourth stage consists of reusable tools and templates. The fifth stage consists of broader ecosystem assessment and continuous improvement.

This staged approach protects IML from excessive abstraction. It also protects it from premature technological development. A tool should be built only when the problem, users, information flows and governance requirements have been sufficiently clarified.

A future implementation pathway should include the search for an institutional reference environment capable of combining academic research, public health expertise, clinical or microbiological field access, software development capacity and independent evaluation. IML should not become only a conceptual framework. It should evolve toward a governed reference implementation that can test digital health software quality against real health information flows and compare existing solutions through transparent, reproducible and health-oriented criteria.

This pathway should also preserve the identity principle established in Domain 3. IML should avoid proposing a new universal identity number. Instead, future prototypes should test how existing national identifiers, digital identity wallets, trusted brokers, temporary access tokens, QR-based authorisation and patient-mediated consent can support linkage, deduplication and access without unnecessary exposure of sensitive identity data.

10.9 Publication, DOI and open documentation

When sufficiently mature, IML should become publicly citable. A digital object identifier, for example through an open research repository, would allow specific versions to be referenced. A public documentation repository could track changes, host templates and invite contributions.

Publication should not be understood only as dissemination. It is part of scientific accountability. A citable version requires clear authorship, transparent methodology, documented limitations and openness to critique. If IML aims to support health information ecosystems, its own documentation should model the transparency it recommends.

10.10 Building an IML community

No framework can mature without a community. IML should progressively invite participation from clinicians, researchers, informaticians, public health authorities, payer organisations, software developers, patient representatives and institutions. The community should not be built around agreement alone. It should be built around constructive disagreement, shared learning and practical improvement.

Community building may begin modestly: a small group of reviewers, a public glossary, an initial AMR demonstrator, a repository of indicators and a set of open questions. Over time, this can grow into working groups, country profiles, implementation pilots and scientific publications.

10.11 Future research agenda

Several research questions emerge from this manuscript. How should Health Information Ecosystems be measured? How can Institutional Engagement be operationalised as an indicator? How can payer interoperability be evaluated across different financing models? How should clinical context preservation be implemented technically and governed ethically? How can AI readiness be assessed without encouraging premature automation?

These questions are not weaknesses of IML. They are its research programme. A mature framework should identify what remains uncertain and organise future work around those uncertainties.

How can digital health software quality be evaluated independently in a market where certification, procurement, adoption or declared functionality may not be sufficient to demonstrate preservation of clinical meaning, auditability, correction capacity, portability, interoperability usefulness and patient safety?

How can identity-light mechanisms such as temporary access tokens, QR-based authorisation, patient-mediated consent, trusted brokers, contextual pseudonyms and episode-based linkage be validated without creating a new identity number or increasing risks of exclusion, surveillance or misuse?

How can professional role alignment be assessed without reducing the question to professional conflict? Which indicators can show whether training, access rights, remuneration, responsibility and real contribution to information quality are aligned across doctors, nurses, pharmacists, laboratory professionals, allied health professionals, public health workers, data professionals and administrative actors?

How can institutional responsiveness be measured as part of interoperability maturity? Which response timelines, contact pathways, review mechanisms and decision-accountability procedures are sufficient for ecosystems that must evaluate digital health innovation safely and quickly?

10.12 From roadmap to shared responsibility

The roadmap proposed here returns to the founding principle of the manuscript: health information is a shared responsibility. Every actor in the ecosystem contributes to the quality, continuity and trustworthiness of information. Every actor can also contribute to its fragmentation.

IML therefore cannot be built by one institution alone. It requires scientific discipline, technical openness, professional engagement, institutional courage and public trust. The roadmap is not a promise that interoperability will be achieved. It is a proposal for how progress can be pursued with humility and persistence.

The future of health information interoperability will not be determined by the number of systems connected, but by the ability of ecosystems to transform trustworthy information into knowledge that protects health. This is the ambition of the Interoperability Maturity Lab.

Key Messages - Part X

- The IML roadmap transforms the manuscript from a conceptual framework into a programme of work.
- Versioning is part of scientific discipline and protects editorial continuity.
- External review should be built into the framework rather than added at the end.
- The methodology handbook and technical handbook should remain distinct but connected.
- Case studies are laboratories for the framework and should begin with AMR/BMR.
- Publication, DOI and open documentation can make IML citable, transparent and reusable.
- Community building should invite critique as well as collaboration.
- Future research questions are part of the value of IML, not signs of incompleteness.
- The roadmap should remain progressive, adaptable and oriented toward health improvement.
- Future IML work should evaluate professional role alignment and institutional responsiveness as practical conditions for digital health software quality and ecosystem learning.

ANNEXES

The following annexes are designed as operational tools for the IML Founding Manuscript. They provide definitions, templates, indicators and working instruments that will evolve through scientific review, implementation experience and future versions of the IML framework.

Annex A - Glossary of Core Concepts

This glossary defines the core vocabulary used throughout the manuscript. The purpose is not to impose fixed terminology, but to provide a stable working language for scientific review and practical implementation.

Concept	Working definition
Health Information Ecosystem	The dynamic network of individuals, organisations, technologies, governance mechanisms and information flows that collectively generate, validate, exchange, interpret and use trustworthy health information in order to protect and improve health.
Continuity of Health Information	The capacity of health information to accompany the patient and the health system across time, settings and authorised actors without losing meaning, integrity or clinical relevance.
Knowledge Continuity	The capacity of successive clinical, public health and research activities to build upon previous validated knowledge rather than repeatedly restarting from fragmented information.
Clinical Context Preservation	The preservation of the clinical circumstances that give health information its meaning, including symptoms, diagnosis, indication, treatment, evolution and relevant risk factors.
Institutional Engagement	The willingness and capacity of organisations to participate in dialogue, cooperation, governance and implementation activities that support interoperability.
Institutional Readiness	The practical preparedness of an organisation or ecosystem to collaborate, respond, align incentives, allocate resources and participate in interoperable health information processes.
Payer Interoperability	The capacity of public insurers, national health services, private insurers and mutual organisations to exchange appropriate information with clinical and public health actors under trustworthy governance.
Open Clinical Workspace	A vendor-neutral, open-source-oriented clinical environment designed to interact with existing systems, import information on demand and preserve clinical context without replacing local software.
Information Stewardship	The collective responsibility to maintain the quality, security, traceability, proportionality and usefulness of health information throughout its lifecycle.
Progressive Interoperability	The principle that interoperability should be improved step by step according to context, resources, priorities and measurable progress rather than through a single ideal model.
Database Quality in Health Information Ecosystems	The capacity of database technologies and data stores to preserve trustworthy health information through integrity, security, traceability, reversibility, portability, correction, backup, restoration and long-term maintainability.

Concept	Working definition
Identity Access Layer	A secure trust and access layer that works with existing national, local or organisational identity systems to generate purpose-limited, auditable and privacy-preserving access tokens without creating a new personal identity number.
Contextual Access Token	A temporary, signed, auditable and revocable proof that identity or authorisation has been verified for a defined health information purpose, under defined rules and for a defined duration.
QR-based Authorisation	The use of a QR code as a practical carrier for a temporary access token or verification link. In IML, a QR code should not contain sensitive identity or health information in clear text.
Digital Health Software Quality Benchmark	A transparent, reproducible and health-oriented evaluation approach for assessing whether digital health software preserves meaning, clinical context, auditability, correction capacity, portability, security, interoperability usefulness and measurable contribution to Health Information Ecosystems.
Professional Role Alignment	The degree to which health professions' training, legal scope, access rights, responsibilities, remuneration, recognition and participation in governance correspond to their real contribution to health information generation, interpretation, correction, reuse and patient communication.
Digital Health Workforce Transformation	The adaptation of health professional education, continuing training, skill mix, teamwork and digital competencies to the information-intensive, interoperable and AI-enabled conditions of contemporary healthcare.
Institutional Responsiveness	The capacity of institutions to receive, acknowledge, review, redirect or decline external scientific, technical or operational proposals through transparent contact points, procedures and reasonable timelines.

Annex B - Acronyms and Abbreviations

Acronym	Meaning
AI	Artificial Intelligence
AMR	Antimicrobial Resistance
BMR	Bacteries multiresistantes / Multidrug-resistant bacteria
DICOM	Digital Imaging and Communications in Medicine
EHR	Electronic Health Record
FHIR	Fast Healthcare Interoperability Resources
HL7	Health Level Seven International
IHE	Integrating the Healthcare Enterprise
IML	Interoperability Maturity Lab
LOINC	Logical Observation Identifiers Names and Codes
MDRO	Multidrug-resistant organism
OECD	Organisation for Economic Co-operation and Development
SNOMED CT	Systematized Nomenclature of Medicine - Clinical Terms
UTI	Urinary Tract Infection
WHO	World Health Organization

Annex C - Proposed IML Indicator Catalogue

The following indicators are preliminary. They are intended to support expert review and future methodological refinement. Each indicator should eventually be linked to evidence sources, scoring guidance and validation procedures.

ID	Domain	Draft indicator
GOV-1	Governance and Standards	Existence of a national or regional interoperability governance structure with defined responsibilities.
GOV-2	Governance and Standards	Publication of implementation guides for core health information exchange processes.
GOV-3	Governance and Standards	Transparent processes for updating standards, terminology and governance rules.
TEC-1	Technical Interoperability	Availability of secure APIs or exchange services for priority clinical information.
TEC-2	Technical Interoperability	Ability to exchange laboratory, prescription, discharge and imaging metadata using recognised standards.
TEC-3	Technical Interoperability	Existence of testing environments, sandboxes or conformance mechanisms.
IDT-1	Identity, Consent and Trust	Reliable patient identification across participating organisations.
IDT-2	Identity, Consent and Trust	Operational consent or lawful-basis mechanisms appropriate to data use.
ADP-1	Adoption and Use	Routine use of interoperable workflows by clinicians rather than pilot-only implementation.
ADP-2	Adoption and Use	Training and support mechanisms for professionals and patients.
SEC-1	Security and Resilience	Defined cybersecurity and incident response procedures for interoperable exchanges.
SEC-2	Security and Resilience	Auditability and traceability of access to sensitive health information.
LRN-1	Feedback, Correction and Learning	Mechanisms for correcting erroneous information across systems.
LRN-2	Feedback, Correction and Learning	Use of interoperability indicators to support continuous improvement.
INS-1	Institutional Engagement	Identified institutional contact points for scientific and operational collaboration.
PAY-1	Payer Interoperability	Defined pathways for appropriate information exchange between clinical actors and payers.
AI-1	AI Readiness	Documented governance for AI-assisted use of health information under human responsibility.
TEC-4	Technical Interoperability	Operating environments are documented and assessed for lifecycle support, portability, maintainability and capacity to support secure interoperable exchange.
SEC-3	Security and Resilience	Operating systems used in clinical or infrastructure settings are covered by patch management, hardening baselines, audit mechanisms and recovery procedures.
TEC-5	Technical Interoperability	The ecosystem documents database technologies used for priority health information, including data models, export formats, migration procedures, backup and recovery mechanisms, audit capabilities and long-term maintainability.
SEC-4	Security and Resilience	Databases storing sensitive health information implement access control, encryption where appropriate, audit logs, role separation, backup testing, recovery procedures and secure configuration aligned with recognised benchmarks.

ID	Domain	Draft indicator
LRN-3	Feedback, Correction and Learning	The ecosystem supports correction, versioning and traceable reuse of database-held information to preserve knowledge continuity over time.
IDT-3	Identity, Consent and Trust	The ecosystem can use existing national, local or organisational identity systems through a secure identity-access layer that generates purpose-limited, auditable and privacy-preserving access tokens without creating a new personal identity number.
IDT-4	Identity, Consent and Trust	QR codes, mobile applications or digital wallets used for access contain temporary, signed and revocable tokens or verification links rather than sensitive identity or health information in clear text.
TEC-6	Technical Interoperability	The ecosystem can test digital health software against transparent and reproducible criteria, including preservation of meaning, clinical context, auditability, correction, portability, reversibility, security, interoperability usefulness and avoidance of vendor lock-in.
LRN-4	Feedback, Correction and Learning	The ecosystem uses software evaluation results, incidents, user feedback, correction requests and interoperability failures to improve software configuration, governance, procurement and implementation over time.
ADP-3	Adoption and Use	The ecosystem documents whether health professions have training, access rights, responsibilities and governance participation aligned with their real contribution to health information generation, interpretation, correction, reuse and patient communication.
INS-9	Institutional Engagement	The ecosystem has documented procedures for receiving, acknowledging, reviewing, redirecting or declining external scientific, technical or operational proposals related to digital health interoperability, with clear contact points and reasonable response timelines.

Annex D - Evidence Grading Template

Evidence grading is intended to distinguish documented facts from expert judgement and IML proposals. This distinction is essential for scientific transparency.

Level	Category	Description
Level A	Direct evidence	Official documents, legal texts, standards, peer-reviewed studies or validated public datasets.
Level B	Documented implementation	Public reports, implementation guides, official dashboards, technical documentation or audited programme information.
Level C	Expert-informed evidence	Structured expert interviews, Delphi processes, professional consensus or institutional responses.
Level D	Exploratory evidence	Preliminary observations, pilots, non-systematic documentation or early-stage project information.
Level E	Research hypothesis	Conceptual proposal requiring validation before use in scoring or comparison.

Annex E - AMR / BMR Minimum Dataset Template

The AMR demonstrator requires clinical meaning. The following dataset is proposed for the MDR E. coli urinary tract infection use case. It is designed to distinguish resistant isolates from clinically meaningful infections.

Dataset block	Elements
Patient context	Age group, sex, pregnancy status when relevant, care setting, comorbidity context.
Clinical symptoms	Dysuria, frequency, urgency, flank pain, pelvic pain, delirium or systemic symptoms when relevant.
Fever and severity	Temperature, sepsis criteria if applicable, hospitalisation status.
Complication context	Male sex, pregnancy, catheter, urinary tract abnormality, immunosuppression, recurrent infection.
Urinalysis	Leukocyturia, nitrites, haematuria when available.
Culture	Specimen type, collection date, organism, quantitative count, contamination markers.
Antibiogram	Susceptibility profile, resistance phenotype, MDR status, ESBL/carbapenemase markers if available.
Clinical interpretation	Colonisation, contamination, asymptomatic bacteriuria, symptomatic infection, uncertain status.
Diagnosis retained	Clinician diagnosis, coding, differential diagnosis where relevant.
Treatment	Antibiotic prescribed, dose, duration, route, treatment modification after results.
Outcome	Clinical improvement, recurrence, hospitalisation, complications, mortality if applicable.
Public health use	Aggregated surveillance, outbreak detection, resistance trend analysis, stewardship feedback.

Annex F - Payer Interoperability Indicators

Payer interoperability is proposed as a distinctive IML contribution. These indicators should be adapted to national contexts, including single-payer, national health service, mixed public-private and mutual insurance systems.

ID	Indicator	Draft description
PAY-1	Mapping of payer actors	Public insurers, national health services, private insurers and mutual organisations are identified as ecosystem actors.
PAY-2	Care pathway continuity	Payer-held information can support longitudinal understanding of care pathways under appropriate governance.
PAY-3	Prevention support	Payer information is used to support prevention, screening or chronic disease follow-up without inappropriate surveillance.
PAY-4	Clinical-payer boundary	Clear rules define what information may be exchanged between clinical and payer systems.
PAY-5	Public-private coordination	Mechanisms exist to coordinate information across public and complementary/private payers where relevant.
PAY-6	Equity monitoring	Payer information contributes to identifying inequalities in access, delays or unmet needs.
PAY-7	Patient transparency	Patients can understand how payer-held information is used and by whom.
PAY-8	Auditability	Access and reuse of payer information are traceable and accountable.

Annex G - Institutional Engagement and Readiness Indicators

ID	Indicator	Draft description
INS-1	Public contact points	Institutions publish or identify appropriate contact points for interoperability, research and collaboration.
INS-2	Responsiveness	Institutions have procedures for responding to external scientific or operational collaboration requests.
INS-3	Collaborative governance	Stakeholders from clinical, technical, patient, payer, public health and research communities participate in governance.
INS-4	Pilot participation	Institutions can participate in limited, well-governed pilot projects without requiring full system replacement.
INS-5	Transparency	Methodologies, results and limitations of interoperability programmes are publicly documented where possible.
INS-6	Learning capacity	Feedback from failures, delays and non-adoption is used to improve implementation strategy.
INS-7	Resource alignment	Financial, human and technical resources are aligned with interoperability objectives.
INS-8	Sustainability	Institutional commitment extends beyond individual projects or short-term funding cycles.
INS-9	Institutional responsiveness to external proposals	Institutions have documented procedures for receiving, acknowledging, reviewing, redirecting or declining external scientific, technical or operational proposals related to digital health interoperability, with clear contact points and reasonable response timelines.

Annex H - Targeted Literature Support for IML Original Concepts

This annex adds targeted literature support for concepts that are central or relatively original within IML. It does not transform IML proposals into established definitions. It maps each concept to adjacent scientific, institutional or standards-based literature that can support external review, future citation work and methodological validation.

The objective is deliberately modest: to show that IML is not built in isolation, while preserving the manuscript's own contribution. A full systematic review should be developed in a later methodology handbook or companion paper.

H.1 Concept-to-literature mapping

IML concept	Targeted literature support	How it supports IML	Remaining validation need
Health Information Ecosystems	WHO Global Strategy on Digital Health 2020-2025; Global Digital Health Monitor; OECD health data governance work; ISO/IEC/IEEE 42010 architecture-description principles.	These sources support the idea that digital health must be understood as a governed ecosystem of actors, services, infrastructure, standards, workforce, data flows and accountability rather than as isolated applications.	IML should later define measurable ecosystem boundaries and relationship indicators for specific use cases.
Knowledge Continuity	Learning Health System literature from AHRQ and the National Academy of Medicine; Haggerty et al. on continuity of care and informational continuity.	These sources support the movement from isolated data to learning, practice improvement and continuity of usable information across time and settings.	IML should specify indicators showing whether knowledge is preserved across encounters, organisations, generations of systems and authorised reuse.
Institutional Engagement	WHO Digital Implementation Investment Guide; CFIR 2.0; Proctor implementation outcomes taxonomy.	These sources support the view that digital health implementation depends on context, readiness, adoption, feasibility, sustainability, organisational setting and implementation process, not technology alone.	IML should validate observable indicators of institutional responsiveness, contact points, participation in pilots and external scientific review.

Payer Interoperability	CMS interoperability and prior authorization rules; HL7 Da Vinci Payer Data Exchange; OECD health data governance.	These sources support the relevance of payer-held information, payer-to-payer exchange, patient access APIs, provenance, governance and links between clinical, administrative and financing information.	IML should adapt payer interoperability indicators beyond the United States to national health services, single-payer systems, mixed public-private systems and mutual insurance contexts.
AI Readiness	WHO ethics and governance guidance for AI in health; NIST AI Risk Management Framework 1.0; WHO guidance on large multimodal models in health.	These sources support the need for human responsibility, transparency, bias monitoring, privacy, safety, accountability, explainability, risk management and post-deployment governance.	IML should test whether AI readiness indicators are linked to information quality, clinical context preservation and real workflow accountability.
Operating System Quality	ISO/IEC 25010:2023; POSIX.1-2024; NIST Cybersecurity Framework 2.0; CIS Benchmarks.	These sources support vendor-neutral assessment through quality characteristics, portability, lifecycle support, cybersecurity risk management, secure configuration and maintainability.	IML should develop a protocol for comparing operating environments in real clinical, laboratory and public health workflows without turning the manuscript into a software catalogue.
Database Quality	ISO/IEC 25010:2023; ISO/IEC 9075:2023 SQL standard; NIST SP 800-53 Rev. 5; CIS Benchmarks; OWASP database security guidance.	These sources support assessment of data stores through integrity, security, auditability, portability, reversibility, maintainability, backup, recovery and secure configuration.	IML should validate database-quality criteria against concrete health datasets, beginning with the AMR/BMR minimum dataset and the Open Clinical Workspace prototype.
Open-source reference implementation and digital health software quality benchmark	WHO Global Strategy on Digital Health; NICE Evidence Standards Framework for Digital Health Technologies; Learning Health Systems; academic medicine and academic health systems; European Health Data Space; Software as a Medical Device clinical evaluation frameworks.	Supports the IML position that the framework should evolve beyond a repository of ideas toward a university-governed, vendor-neutral, open-source software reference capable of testing IML indicators, datasets and interoperability methods against real health information flows and comparing them with market solutions.	Requires identification of an academic or public health partner, concrete field of experimentation, software governance model, open-source maintenance strategy, cybersecurity safeguards, legal and ethical framework, data hosting model, access rules, correction procedures, benchmarking protocol, clinical safety indicators, comparison with market solutions and sustainability plan.
Identity access layer and trusted digital identity	European Digital Identity Wallet; Regulation (EU) 2024/1183 establishing the European Digital Identity Framework; European Health Data Space; Regulation (EU) 2025/327 on the European Health Data Space.	Supports the IML position that it should not create a new personal identity number, but explore secure, purpose-limited and auditable access mechanisms compatible with existing national and European digital identity infrastructures.	Requires validation of legal basis, privacy impact, cybersecurity, token design, consent, auditability, revocation, correction procedures, biometric safeguards, exclusion risks, interoperability with national eID systems, and alignment with EHDS governance and national health data laws.
Professional role alignment and digital health workforce	Frenk et al. on health professional education reform; WHO Global	Supports the IML position that interoperability maturity depends not only	Requires country-specific validation of scopes of practice, training

transformation	Strategy on Human Resources for Health: Workforce 2030; WHO Global Strategy on Digital Health 2020-2025; OECD work on skill mix and policy change in the health workforce; Learning Health Systems.	on systems and standards, but also on whether professional training, roles, responsibilities, access rights, recognition and remuneration match the real work of information-intensive healthcare.	curricula, remuneration structures, legal responsibilities, digital skills, professional hierarchy, patient safety impact, acceptability among professions and measurable effects on information quality.
Institutional responsiveness and governance adaptation	Implementation science; Learning Health Systems; WHO digital health governance; public-sector innovation and institutional capacity literature.	Supports the IML position that interoperability maturity depends not only on infrastructure and standards, but also on whether institutions can respond to external proposals, evaluate innovation, adapt governance and connect ideas to experimentation.	Requires validation of response pathways, contact points, review mechanisms, administrative timelines, decision accountability, openness to academic and open-source models, and ability to link scientific proposals to operational testing.

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H.3 Editorial note

The references listed here are targeted supports, not an exhaustive literature review. Before formal publication, IML should separate three layers of evidence: sources supporting the conceptual vocabulary, sources supporting methodological indicators, and sources validating operational performance in concrete use cases.

End of targeted literature support annex

These annexes remain working instruments for review. They are provided for discussion, refinement and future validation rather than as final methodology.

Annex I – Existing Open-Source Building Blocks Relevant to the Open Clinical Workspace
Purpose of this annex

This annex identifies existing open-source digital health platforms, architectural frameworks and technical components that could be evaluated, reused or connected within the Open Clinical Workspace.

Inclusion in this annex does not constitute endorsement. Each component should be assessed according to IML criteria, including clinical usefulness, interoperability, security, auditability, correction, portability, accessibility, offline capability, governance, community sustainability and total cost of implementation.

Component	Current role and availability	Potential contribution to IML
OpenMRS	Mature open-source clinical record platform. OpenMRS reports use in more than 8,100 sites across over 80 countries, supporting more than 22 million patients.	Possible clinical-record and patient-management foundation. IML could add AMR workflows, clinical-context preservation, cross-system correction and evidence grading.
Bahmni	Integrated open-source hospital and EMR system designed for low-resource settings. It combines OpenMRS, laboratory, imaging, inventory and administrative components. It can be hosted locally without dependence on permanent Internet access and reports more than 500 sites in over 50 countries.	Strong candidate for institutions requiring an immediately usable hospital system. It could provide a low-resource deployment profile for the Open Clinical Workspace rather than requiring IML to recreate an entire hospital system.
DHIS2	Mature open-source platform for health information management, surveillance, programme monitoring and analysis. It is used by health ministries in 80 low- and middle-income countries; 3.2 billion people live in countries where it is deployed.	Public health reporting, aggregated surveillance, dashboards and population-level analysis. Particularly relevant for sending privacy-preserving AMR indicators from clinical services to public health authorities.
OpenELIS Global	Open-source laboratory information system designed for individual laboratories and national laboratory networks. It reports more than 1,000 deployed laboratories, 18.7 million patients supported and use in more than 26 countries. Its architecture supports FHIR	Principal candidate for the laboratory component of the AMR/BMR demonstrator, including orders, microbiology results, susceptibility testing

Component	Current role and availability	Potential contribution to IML
	and offline operation.	and exchange with clinical systems.
OpenHIE	Open architectural framework and community for national and regional health information exchange. OpenHIE is not a single downloadable application; it provides reusable architectural patterns, workflows and component definitions.	Architectural reference for linking patient, provider, facility, terminology, laboratory, clinical and public health services without imposing a single national software product.
OpenHIM	Available open-source interoperability middleware. It supports secure communications, routing, orchestration, message transformation, transaction logging and monitoring between heterogeneous systems.	Possible interoperability gateway for connectors between the Open Clinical Workspace, OpenMRS, OpenELIS, DHIS2, payer systems and national infrastructures.
HAPI FHIR	Active open-source implementation of the HL7 FHIR standard, including client libraries, parsers, validation tools and deployable FHIR servers.	Possible FHIR repository and API layer for standardised exchange, validation, subscriptions and integration with external systems.
Open Concept Lab	Open-source terminology-management infrastructure supporting concept dictionaries, mappings, code systems and FHIR-enabled terminology services. It is being used in national-scale terminology work, including current collaboration in Brazil.	Terminology governance, mapping of local codes, ValueSet management and semantic consistency for microbiology, diagnoses, medicines and clinical observations.
Community Health Toolkit	Open-source toolkit for digital community-health applications, designed for difficult-to-reach environments and offline-first use. It supports smartphones, basic devices, messaging, tasks, longitudinal profiles and decision-support workflows.	Community care, home follow-up, prevention and low-connectivity workflows outside hospitals. It could link community health workers with the Open Clinical Workspace.
OpenSRP 2	Open-source, FHIR-native platform for frontline health workers, community programmes and health facilities. It supports offline data collection and low-connectivity operation.	Mobile registration, community follow-up, vaccination, maternal health and other future IML use cases requiring offline longitudinal records.
Keycloak	General-purpose open-source identity and access-management system supporting authentication, user federation and authorisation.	Possible technical component for professional authentication, organisational identities, role-based access and federation. It would not replace national patient identity systems.
PostgreSQL	Mature open-source relational database with long-term community development and support for reliable, structured data workloads.	A practical initial database for prototypes, audit logs, clinical-context relationships and AMR datasets, provided that portability, backup, correction, migration and reversibility remain explicitly tested.

Annex conclusion

No single existing project currently provides the complete IML vision. Several already solve substantial parts of the problem, however, and together they form a powerful foundation.

The likely IML strategy should therefore be **composition rather than duplication**:

- OpenMRS or Bahmni for clinical care;
- OpenELIS Global for laboratory workflows;
- DHIS2 for surveillance and public health;
- OpenHIE and OpenHIM for architecture and exchange;
- HAPI FHIR for standards-based APIs;
- Open Concept Lab for semantic interoperability;
- Community Health Toolkit or OpenSRP for offline and community use;
- Keycloak for professional access;
- PostgreSQL for an initial portable data foundation.

IML would supply the missing connective tissue: clinical context, correction, evidence grading, software-quality evaluation, institutional and payer interoperability, and the continuous transformation of information into trustworthy knowledge.

Annex J - Suggested Citation and Review Template

Suggested citation

Barrios R. Interoperability Maturity Lab (IML): A Scientific Framework for Trusted Health Information Ecosystems. Founding Manuscript. Current working draft. Open for Scientific Review. 2026.